

Oracle Cloud Capacity Analytics Sizing User's Guide

Exadata On-Premises

Exadata Database Service on Dedicated Infrastructure

Exadata Database Service on Cloud@Customer

Base Database Service

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Table of contents

Release Highlights	6
Introducing OCCA	6
Sizing Disclaimer	6
Sizing Application Components	6
De-Support of EMCC/BAR	7
Sizing Overview	7
Sizing Methodologies	7
The Predictive Sizing Methodology	7
The Comparative Sizing Methodology	7
Database Resource Consumption Model	7
DB CPU Consumption in AWR Reports	8
Host CPU Utilization in AWR Reports	8
Plotting DB CPU Consumption vs. Host CPU Utilization	8
Understanding vCPU, OCPU, and ECPU Metrics	9
Oracle Compute Unit (OCPU)	9
Elastic Compute Unit (ECPU)	9
Over-Provisioning	9
Automatic Handling of Outlier vCPU Values	10
Comparative Sizing Data Collection	10
Inventory	10
Consumption	10
Utilization of System Resources	10
OCCA Sizing Data Collection	10
Understanding Adjusted Hourly Metrics	11
Hourly Metrics	11
Daily Metrics	11
Adjusted Hourly Metrics	11
Adjustment Plots	12
Cohorts: Grouping Databases for Consolidation	12
Isolation Requirements	12
Cohort Isolation Example	13
Impact of Consolidation vs. Isolation	13
Individual Database Metrics vs. Cohort Metrics	13
Handling Missing Metrics	14
Old Enterprise Manager Server Versions	14
Old Enterprise Manager Agent Versions	14
Enterprise Manager Job Failures	14
Enterprise Manager Job Queue Processes Disabled	14
Targets in Bad Status	15

Stale Metrics	15
Metric Collection Disabled	15
Standby Databases	15
Errors During Extract Process	15
Correcting Missing Metric Issues	16
Determining Values to Substitute for Missing Metrics	16
Missing Instance Metrics	16
Understanding Data Guard Workloads	17
Missing Metrics for Standby Databases	17
Storage Sizing for Standby	17
Physical Compute Shape Sizing	17
vCPU/OCPU for Normal Operation	18
vCPU/OCPU for Disaster Operation	18
OCCA Controls for Copy DB Metrics	18
Directive: Physical Standby	18
Directive: Primary + Standby Instance Metrics	18
Directive: Standby Instance Metrics	19
Directive: All	19
Adjustment Plots for DR Workloads	19
Reduce DB Metrics	20
Sizing, Capacity Planning, Consumption, and Performance	20
Sizing	20
Resource Consumption	20
Capacity Planning	21
Performance Tuning	21
Understanding Exadata Performance Benefits	21
Percentile Calculations	21
Sizing Data Collection	22
OCCA AWR Miner	22
EMCC Extract Scripts (Inventory & Resource Consumption)	22
Manual Input	22
EMCC Extract Scripts	22
Versions Supported	22
SQL Plus Version Requirement	22
EMCC & Pack License Requirements	22
Oracle Employee Laptop Setup	22
Keep OCCA Package Updated	23
OCCA for EMCC Extracts - Directory Structure	23
Customer Scripts	24
Source Side Filtering	24
Source Side Obfuscation	24
Tool Demo Data	24

OCCA Sizing Process	24
Process Overview	24
Step #0: EMCC Health Assessment	25
Pre-Flight Check	25
Step #1a: Customer Executes EMCC Extracts	25
Step #1b Optional: Source Side Filter	26
Step #1c Optional: Source Side Obfuscation	26
Step #2: Create Property Files	27
Step #3: Data Guard Metrics	27
Step #4: Edit Properties	27
How to Exclude Databases	28
Copying Properties from EMCC Extracts	28
Step #5: Add Properties	28
Step #6: Run Metric Analysis	28
Step #7: Review Audit Information	28
Step #8: Upload and Process Opportunity in OCCA	29
Step #9: Produce Sizing Report	29
Flash Cache Sizing	29
DB Segment I/O Analysis Limitations	30
Allocated vs. Used vs. Working Set	30
Importance of Flash Cache Sizing	30
Recommendations for Sizing Reports	31
Recommended Contents	31
Summary of Missing Metrics	31
OCCA Reports	32
OCPU Sizing (Cloud)	32
OCPU Peak for Standby with OS+GI	33
OCPU Peak with OS+GI	33
Combining Cohorts Using VM Clusters	33
Sub-Dividing Large Cohorts	33
Resource Consumption Plots	33
Annual Adjustment & Seasonality	34
Resource Consumption Percentiles	34
Adjusted vCPU Consumption	34
Adjustments & Seasonality	34
OCPU Management Plan	35
Growth & Fault Tolerance	35
Adding Capacity for Cohorts	36
OCCA Desktop Analytics Parameter Settings	36
Using OCCA AWR Miner	37
OCCA AWR Miner Components	37
OCCA AWR Miner Sizing Steps	38

Customer Executes OCCA AWR Miner Scripts	38
AWR Snapshot Intervals & Retention	38
Time Zone Handling	39
AWR Snapshot Overlap	39
OCCA AWR Miner Cohorts	39
OCCA AWR Miner Database Plots	40
Handling Non-US Numeric Separators	40
OCCA Web Manual Data Entry	40
Sum of Peaks Method	41
Debugging OCCA Issues	41
Collecting Sizing Artifacts	41
Debugging for Data Guard Handling	41
Evaluating Server Utilization vs. DB Consumption	41
Evaluating Attenuated DB vCPU Outliers	41
Automatic Elimination of CPU Outliers	42
OCCA Command Reference	43
Conclusion	43
References	44
OCCA User's Guide for Oracle Employees	44
EMCC Extracts Guide for Customers	44
Scripts for Customer Extract	44
OCCA Analytics Desktop Package for Oracle Employees	44
OCCA Test Data (EMCC Extract)	44
OCCA Web User Interface	44
Introduction to OCCA Presentation	44
Pricing Calculators	44

Release Highlights

This release of Oracle Cloud Capacity Analytics (OCCA) User's Guide includes important information for using OCCA including:

1. Disclaimer for all Customer Presentations
2. New Guidance on Causes of Missing Metrics
3. Flash Cache Sizing
4. Desupport of backward compatibility with BAR Tool

Introducing OCCA

Oracle Cloud Capacity Analytics (OCCA) uses Data Analytics for capacity sizing for both Cloud and on-premises environments. This version of the OCCA includes important enhancements that make the sizing process simpler to execute including:

1. Users arrange databases into "cohorts" for consolidation
2. OCCA is database-centric rather than server-centric
3. OCCA is based on consumption rather than utilization
4. OCCA uses time-aligned peaks rather than sum of peaks
5. OCCA uses fine-grained hourly data combined with 1 year of daily history

Sizing Disclaimer

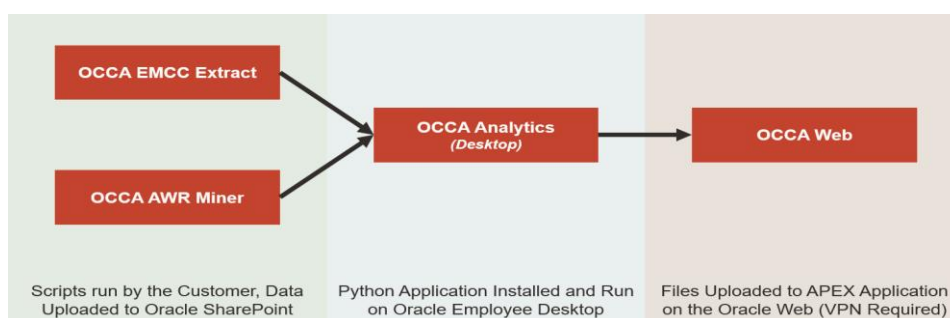
All customer presentations should include the standard disclaimer language, which highlights customer involvement in the process and the inherent limitations of any sizing engagement.

"Hardware and software sizing and related information contained herein is an estimate provided for discussion purposes only, based on information provided by you, and shall not be construed as any representation or warranty, express or implied, on behalf of Oracle, including any warranties or conditions of merchantability and fitness for a particular purpose."

Customers provide input to the sizing process and the result is highly dependent on that input. The sizing process does not include any future growth projections, which are only known to the customer. There can also be significant delay between the sizing process and eventual system implementation and there can be growth that occurs during that period as well. Changes in application behavior, application version upgrades, and database system changes can also impact sizing requirements that are not accounted for in the sizing process.

Sizing Application Components

There are several components that are used for system capacity sizing including EMCC Extracts, AWR Miner Extracts, OCCA Analytics, and OCCA Web as follows:



EMCC Extracts and OCCA AWR Miner scripts are run by customers, producing data that is fed into OCCA Desktop Analytics to perform analysis of source database resource consumption. Data from OCCA

Analytics is fed into OCCA Web, where target selection is performed. It is important to note that there is a specific version of AWR Miner designed for use with OCCA. Other versions of AWR Miner are not compatible.

De-Support of EMCC/BAR

As of the latest release OCCA no longer supports backward compatibility with the BAR sizing tool. BAR and OCCA use different processes for handling sizing and will not necessarily produce the same results in all cases.

Sizing Overview

Sizing is a collaborative effort between customer and Oracle. Accurate sizing relies on accurate data including an accurate inventory of systems and databases, as well as accurate resource consumption metrics. The EMCC extract method allows us to gather the entire inventory of systems and databases reflected in an EM Repository in a single extract operation. This method also allows us to pull up to 12 months of historical resource consumption metrics, far exceeding what's typically available using tools such as AWR Miner.

Sizing Methodologies

Sizing can be done using two different methodologies depending on the situation. These sizing methodologies include the following:

- Predictive Sizing
- Comparative Sizing

The Predictive Sizing Methodology is used for NEW application deployments, such as a new deployment of EBS, SAP, or other packaged application. Many application vendors publish sizing guides, which are based on key metrics of the business application that influence system sizing. These metrics are specific to the application, such as number of inventory items for an Inventory Management application, or GL Code Combinations for a General Ledger application.

The Comparative Sizing Methodology sizes a replacement system based on a comparison to an existing system configuration and resource consumption metrics gathered from that system. The tools and processes outlined in this document implement the Comparative Sizing Methodology.

OCCA supports comparative sizing. For predictive sizing, users should work with application-specific sizing guides and methodologies designed to predict system capacity needs.

Database Resource Consumption Model

OCCA (Oracle Cloud Capacity Analytics) implements the comparative sizing methodology based on database resource consumption. Other sizing processes are typically server-centric, meaning they are based on server configuration and utilization of system resources. OCCA is database-centric, calculating the resources consumed by each database. This new analytic model provides several advantages as follows:

- Eliminates the need for manual server lookup
- Enables changes to database groupings when re-deploying databases
- Enables source-side filtering of database metrics
- Compatible with Cloud deployments where configurations are variable (ex: OCPU)

This model eliminates the need for labor-intensive lookups for each server model and how those servers are configured. It also allows for databases to be easily grouped together in the NEW deployment without forcing the same groupings used on the existing deployment. The database resource consumption model will also enable source-side filtering (in the future) because resource consumption is tied to each database instead of utilization tied to each server. The model is also compatible with Cloud deployments where server (VM) configurations can be variable such as the OCPU setting in Oracle Exadata Cloud Service and Exadata

Cloud@Customer. The previous method of sizing based on percent utilization does not work in Cloud environments that have a variable OCPU configuration.

DB CPU Consumption in AWR Reports

Oracle Database constantly interrogates the O/S for CPU and other resources each instance is consuming. These consumption metrics serve as the basis for the sizing algorithms in OCCA. The database CPU consumption can be seen in standard AWR Reports under the Load Profile section as seen below:

Load Profile				
	Per Second	Per Transaction	Per Exec	Per Call
DB Time(s):	18.6	0.2	0.00	0.00
DB CPU(s):	7.3	0.1	0.00	0.00
Background CPU(s):	0.7	0.0	0.00	0.00
Redo size (bytes):	1,448,836.6	15,415.2		
Logical read (blocks):	1,752,498.0	18,646.1		

Oracle Database uses O/S dependent system calls such as times() or clock() to capture this information, then store the results in AWR and transmits that information to Oracle Enterprise Manager. The DB CPU seconds per second value reflects the number of logical CPUs the Database consumed during a given time period.

Host CPU Utilization in AWR Reports

Oracle Database constantly interrogates the O/S for resource utilization, giving the overall view of how busy the host (or system) was during a given time period. This information is visible in standard AWR reports under the Host CPU section as shown below:

Host CPU								
CPUs	Cores	Sockets	Load Average Begin	Load Average End	%User	%System	%WIO	%Idle
160	40		12.09	26.26	9.5	3.0	0.8	87.5

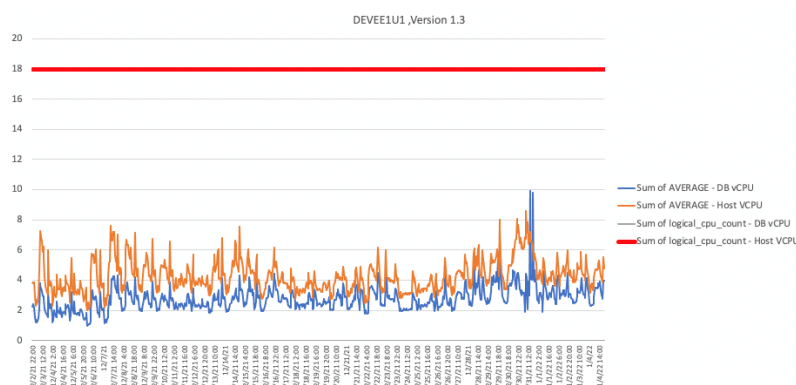
CPU utilization at the host level reflects usage by all databases on a given host as well as any non-database workloads that might also be running on that host. CPU utilization reported as Host CPU in AWR is the same information that would be reported by O/S utilities such as "top" or "vmstat" on Linux.

Plotting DB CPU Consumption vs. Host CPU Utilization

OCCA provides utilities for plotting DB CPU consumption vs. Host CPU Utilization to help users understand the relationship between these metrics. OCCA provides the ability to reduce the volume of data to enable use of tools such as Microsoft Excel.

- Command: **occa --reduce-hourly-metrics**
- File Generated: **sizing/reduced_hourly_metrics.csv**
- Analysis Spreadsheet: **<occa path>/Visualize_reduced_hourly_metrics.xlsx**

The OCCA path can be displayed by running the command "**occa --path**" to locate the spreadsheet distributed for this purpose. The following example is a deviation showing the DB CPU consumption that exceeds CPU utilization of the host, indicating an error in the data:



Logically, the database cannot consume more than the host can utilize, so this condition indicates an anomaly in the underlying data. In most cases, this condition will only affect individual databases or smaller cohorts. However, if this condition is widespread, adjustments may be necessary in the final sizing. As of this writing, OCCA does not automatically perform such adjustments.

Understanding vCPU, OCPU, and ECPU Metrics

Each computing platform has a concept of "virtual" CPU (vCPU) but using different terminology to describe this hardware feature. Each processor "chip" contains multiple processor "cores" and each core will typically contain 2 or more vCPU using the terminology below:

- Intel Hyper-Threading Technology (HT)
- IBM POWER - Simultaneous Multi-Threading (SMT)
- AMD - Simultaneous Multi-Threading (SMT)
- SPARC - Processor Strand
- VMware - Virtual CPU
- Cloud Vendors - vCPU

The most commonly used term is vCPU, so we use this to refer to ALL platforms, regardless of terminology used by a particular vendor. Configuring vCPU does not require over-provisioning, which is a common cause of performance issues due to resource constraints that can result from over-provisioning. The number of vCPU on a processor core will differ according to the manufacturer, and these features can also be enabled or disabled. When these features are disabled, each processor core has a single vCPU. Intel chips currently support 2 Hyper-Threads (2 vCPU) per processor core.

Oracle Compute Unit (OCPU)

An Oracle Compute Unit (OCPU) provides CPU capacity equivalent of one physical core of an X86 (AMD and Intel) processor with hyperthreading enabled. Each OCPU corresponds to two hardware execution threads, known as vCPUs. Database resource consumption is measured in vCPU, OCCA Analytics computes resources in terms of vCPUs, then OCCA Web produces sizing reports in terms of both vCPUs, OCPUs, and ECPUs

Elastic Compute Unit (ECPU)

An ECPU is an abstracted measure of compute resources. ECPUs are based on the number of cores elastically allocated from a pool of compute and storage servers.

Over-Provisioning

Virtualization technology allows administrators to over-provision system resources including vCPU over-provisioning. When a system is over-provisioned, the Virtual Machines on a physical server can be assigned more vCPU than physically exist on a system. For example, a physical server might have 50 processor cores and 100 vCPU available, but an administrator can create 20 Virtual Machines with 10 vCPU each, over-provisioning the hardware by a factor of 2. Customers can over-provision systems at the vCPU level using VMWare, Xen, KVM or other virtualization software. Some Cloud vendors are also known to over-provision systems as well. Oracle Cloud does not over-provision systems, so 1 OCPU equals 2 vCPU in OCI.

Automatic Handling of Outlier vCPU Values

OCCA automatically handles metric collection errors that produce outlier vCPU values that can lead to incorrectly over-sizing systems. These outlier metrics issues occur at a variety of levels in a system, including from the source hardware and operating system. Examples of these issues include the following:

- O/S CPU usage exceeding 100%
- Databases exceeding vCPU entitlement
- Inflated vCPU consumption on multi-threaded systems

Operating systems cannot use more CPU than is available on a system. Databases managed under DBRM can exceed their vCPU entitlement, or approximately 110% of the CPU_COUNT setting. Some multi-threaded systems with high numbers of hyper-threads per core will report inflated vCPU consumption metrics under certain conditions. OCCA automatically detects and corrects for these conditions. Please refer to the section on debugging for more information on how to view detail of outlier handling.

Comparative Sizing Data Collection

Sizing a replacement system based on comparison with existing systems involves collecting information that is broken down into the following categories:

- Inventory of databases and systems
- Consumption of resources by databases
- Utilization of system resources

One of the biggest challenges for data collection is the lack of standardization across the industry in how IT organizations gather and store this information as we shall see later in this document.

Inventory

At the most basic level, an inventory for sizing is a list of databases and the servers they run on. OCCA is designed to collect this information via Oracle Enterprise Manager and OCCA AWR Miner. Prior versions required capture of ALL databases and server mappings to properly account for server resource utilization. OCCA is based on DB consumption of resources, so server resource utilization is not factored into OCCA analytics.

Consumption

OCCA is designed based on database consumption of resources including vCPU, memory, I/O, and storage space. OCCA is compatible with data collected via OCCA AWR Miner, and EMCC extract scripts. Other consumption metric collection methods are not supported with OCCA, but summary information can be entered manually into OCCA Web.

Utilization of System Resources

EMCC Extracts gather system (system or host) resource utilization metrics that are used in limited circumstances in OCCA where attenuation of database CPU metrics are required.

Oracle Enterprise Manager is widely used but is not universally used by all Oracle customers. In addition, AWR is not available in Oracle Database Standard Edition, so the OCCA tool needs to support multiple methods of data collection including automated collection and manual data entry.

OCCA Sizing Data Collection

The OCCA sizing tool supports collection of data using 3 different methods that apply based on how customers manage their systems and customer ability to collect and share data with Oracle. The 3 sources of data are as follows:

- Oracle Enterprise Manager Cloud Control (EMCC) Extract

- Oracle Automatic Workload Repository (AWR) Extract (AWR Miner)
- Other/Manual Data Collection

EMCC Extracts capture structure of targets and relationships, database metrics, and host metrics. EMCC Extracts include information for many databases in a single execution, minimizing work required by the customer. EMCC Extracts include ~33 days of hourly data, as well as 12 months of daily data.

OCCA AWR Miner extracts information from the AWR History tables of a single database. Most customers configure AWR to retain 8-30 days of historical data, as opposed to the 12 months of data available from EMCC. AWR Miner is more time consuming for any customers who have large numbers of databases.

Manual Data collection allows customers to provide data from virtually any source, and data can be manually entered into OCCA. Sales Consultants simply create an opportunity manually in the OCCA Web tool and enter information provided by the customer.

Understanding Adjusted Hourly Metrics

OCCA analyzes EMCC metrics to produce resource consumption data that reflect the fine-grained detail of hourly data, combined with fluctuations in resource consumption that occur day-to-day and through the year.

Hourly Metrics

Hourly metrics include consumption of vCPU, Memory, I/O, and Storage Space for each database during each hour for the past N number of days. The OCCA/EMCC extract process will extract up to 3 months of data if available, but most customer deployments will have approximately 1 month of data (often 33 days).

Daily Metrics

Daily metrics are a rollup of hourly metrics and are stored the Enterprise Manager repository for 2 years. The OCCA/EMCC extract process will extract daily metrics for the preceding 12 months, in 3-month sets identified as, 3 (months 1-3 in the past), 6 (4-6), 9 (7-9), and 12 (10-12), if available. Daily metrics are used to determine peak processing and seasonality of consumption.

Adjusted Hourly Metrics

The analytics processing in OCCA uses BOTH hourly metrics for high fidelity and daily metrics for trend analysis and seasonality. Hourly data gives the necessary detail for accurate sizing combined with the highest peak of resource consumption available in the daily data. The window of hourly data is adjusted UPWARD (only) if a higher consumption peak is present in the daily data. The following example shows hourly vCPU consumption metrics that have been adjusted to a prior peak. The **RED** line on this chart corresponds to the **RED** line on the adjustments plot.

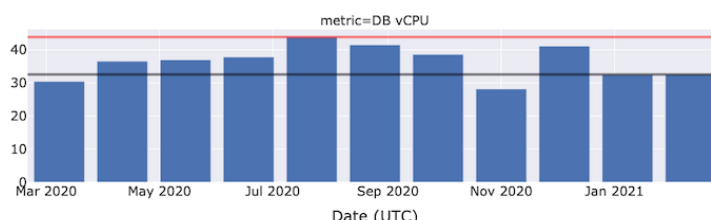


The blue dotted lines on these plots also show reference lines for 95th, 90th, and 80th percentiles as noted in the chart legend. The red line represents a "set and forget" basis for OCPU over the prior 12 months. The black line that conforms to the top of the resource fluctuations is the aggregate resource consumption, while the green line is the average of this black line. The green line is the FLOOR, meaning the lowest possible OCPU charges that could possibly be achieved are based on this value.

If the customers implemented some form of OCPU management, the resulting OCPU charges will be based on a value falling between the red and green lines.

Adjustment Plots

OCCA generates adjustment plots that show how adjustments have been made. These plots are available in 2 versions that show the metric values as well as the relative multipliers between the latest ~33-day period and prior periods within the dataset. The following example shows the adjustment made in the data because of a peak that occurred in a prior period.



Adjustment plots are crucial for understanding the customers' environment and what has transpired during the timeframe under analysis. The above example shows a prior peak processing period occurred during the January/February timeframe as indicated by the **RED** line. The solid **BLACK** line is the peak in the latest ~33-day period.

Cohorts: Grouping Databases for Consolidation

Databases are now grouped into "cohorts" or groups of databases that can be consolidated together in a Bare Metal Server, Bare Metal Cluster, Virtual Machine, or Virtual Machine Cluster. The OCCA tool allows databases to be placed into different cohorts on the target and does not force the same cohort mapping used in the source. Grouping databases into cohorts is driven entirely by the need to isolate databases from each other.

Isolation Requirements

Customers often have one or more reasons why databases need to be isolated from each other. The following table outlines the 6 primary reasons why isolation is required, as well as the underlying drivers behind that isolation.

Physical Location	PROD vs. DR Sites, Location of Users, Integration, Network Configuration
Administrative Separation	Multiple DBA Teams, License Management, DEV/TEST vs. PROD
Security Separation	Quarantine Networks, highly sensitive data (such as PCI, PII), etc.
Blast Radius	Scope of Failure Impact
Maintenance	Database Version Requirements, Upgrade Schedules
Noisy Neighbors	Prevent Performance Issues from impacting other applications

Customers typically have a minimum of two physical locations (two data centers, two Cloud Regions or two Cloud Availability Domains), where one location hosts production systems and the second location hosts disaster recovery. Some customers run both production and disaster recovery systems at BOTH sites in an A/B configuration rather than having designated production vs. disaster recovery locations. Physical location

requirements can also be dictated by "Availability Domain" designations within a Cloud Region, with separate power, cooling, networking, and physical data center (typically 1 kilometer from other ADs within the region). Customers might have other physical isolation requirements such as use of separate physical or virtual networks. Customers are often in the process of closing older locations or opening new locations, so it's important to have discussions about their plans and physical location requirements for databases.

It is critical to understand what databases might be consolidated together onto the same systems, and which ones must reside on separate systems. In addition to physical location requirements, customers might also require administrative separation and security separation. If these first levels of separation don't provide sufficient isolation when failures occur, customers sometimes want to add isolation to limit the "Blast Radius" or scope of failure impacts. Maintenance isolation can normally be handled within Exadata systems using multiple Oracle Homes or multiple Container Databases, and "Noisy Neighbor" isolation can typically be handled using Oracle Resource Manager (DBRM/IORM) features.

Cohort Isolation Example

The customer needs to provide guidance on the cohorts required and how databases are assigned to each of them. The most common configuration uses 4 cohorts, including 2 separate sites with 2 cohorts for production and 2 cohorts for non-production databases as shown below:

- **SITE1-PROD** (PROD and DR)
- **SITE1-NON-PROD** (DEV, TEST, QA, STAGE)
- **SITE2-PROD** (PROD and DR)
- **SITE2-NON-PROD** (DEV, TEST, QA, STAGE)

In the above example, all non-production databases (DEV, TEST, QA, and STAGE) will be isolated from production databases at each location, while "production" includes both PROD and DR databases. Use of dedicated "disaster recovery" sites is becoming less common, with customers using an A/B configuration, where some PROD and some DR databases reside at each site. The test environment(s) should provide the ability to test the DR configuration, so non-production should have some amount of capacity at each site. Virtual Machines (or Clusters of Virtual Machines) can often be used to provide isolation within a physical system if the cohorts are not large enough for dedicated physical systems.

For Cloud deployments, the 2 sites can be in different Cloud Regions, or in 2 different Availability Domains (ADs) within a Cloud Region. Customer requirements regarding distance between production and disaster recovery will dictate whether it's possible to deploy in separate ADs.

Impact of Consolidation vs. Isolation

Consolidating databases results allows them to share resources such as CPU, Memory, Disk Space, and I/O. Combining databases into 1 cohort will typically result in LOWER resource requirements than using 2 cohorts. Conversely, dividing 1 cohort into 2 will typically result in HIGHER resource requirements. In short, the impact is generally as follows:

Number of Cohorts	Isolation	Consolidation	Resource Needs
More Cohorts	Increased Isolation	Decreased Consolidation	Higher resource needs
Fewer Cohorts	Decreased Isolation	Increased Consolidation	Lower resource needs

In addition to the impact on resource needs as more databases are combined into cohorts, there will also be some amount of excess capacity assigned to each cohort. For example, if 1 cohort requires a single Quarter Rack, dividing it into 2 cohorts will require TWO Quarter Racks. This occurs because databases within a cohort share any excess capacity, but those databases can also take advantage of divergent peaks in processing as reflected in the time-aligned plots of resource consumption.

Individual Database Metrics vs. Cohort Metrics

OCCA provides consumption metrics for individual databases as well as cohorts. These metrics will be dramatically different because cohort metrics are time-aligned. Each database will see varying resource consumption throughout a given hour, day, week, and month. When one database is idle, another database might be busy. OCCA analyzes these trends across all databases within a cohort and calculates the resource consumption for the cohort as a whole. Metrics for individual databases are used for targets such as BaseDB that use fully isolated (dedicated) resources for each database. Resources provisioned for each database must be sufficient to handle peak processing loads for that database, even if the database is idle at other times. Cohort metrics are used for targets that allow database consolidation including Exadata, Exadata Cloud@Customer, and Exadata Cloud Service.

Handling Missing Metrics

EMCC extracts frequently have missing metrics that are needed for sizing. The most common causes of missing metrics are:

1. Old Enterprise Manager Server Versions
2. Old Enterprise Manager Agent Versions
3. Enterprise Manager Job Failures
4. Enterprise Manager Job Queue Processes Disabled
5. Targets in Bad Status
6. Stale Metrics
7. Metric Collection Disabled
8. Standby Databases
9. Errors During Extract Process

Old Enterprise Manager Server Versions

OCCA is designed to work with EMCC version 13.x and above. Versions older than 13.x may not necessarily contain the metrics needed for sizing. Customers running older versions should expect missing metrics or other issues and inability to size databases managed and monitored by EMCC. Upgrading an EMCC system will allow collection of metrics moving forward, but it will take 1 year to collect 1 year of new metrics.

Old Enterprise Manager Agent Versions

Upgrading all EMCC agents to version 13.x on all managed targets may also be necessary to collect the information needed for sizing. This upgrade will allow collection of metrics moving forward, but it will take 1 year to collect 1 year of new metrics.

Enterprise Manager Job Failures

Some of the metrics extracted by EMCC are generated by Enterprise Manager jobs that run as part of the normal Enterprise Manager system configuration. In a healthy system, these jobs will execute normally, and the affected metrics will be generated. EMCC Extracts generates the following file to help diagnose problems with Enterprise Manager Jobs:

- `emcc_sizing_em_jobs.csv`

The number of jobs and tasks performed by each may vary depending on the version of Enterprise Manager. Job failure will increment the `FAILURE_COUNT` column, and any non-zero number of failures should be investigated. The `LAST_START_DATE` column should be relatively recent for all of these jobs. In some cases, the number of `JOB_QUEUE_PROCESSES` is either zero or not sufficiently high to execute the jobs in a timely manner.

Enterprise Manager Job Queue Processes Disabled

Enterprise Manager Jobs are executed by Job Queue Processes. An insufficient number of Job Queue Processes will result in delayed processing EM Jobs and may result in missing metrics. Most EMCC systems will have 50 Job Queue Processes defined. Job Queue Processes is set to ZERO is a known problem but check the Last Start and Next Start dates to evaluate.

Targets in Bad Status

Some number of databases, instances, or servers (i.e., "targets") will almost always have missing metrics in Enterprise Manager. Customers often leave decommissioned databases and other targets defined in Enterprise Manager even when systems or databases are no longer operational. Metrics will be missing in these cases and that is a normal, valid condition. Databases in the following statuses will not have associated metrics:

METRIC COLLECTION STATUS	POSSIBLE CAUSES
Blackout	An administrator has disabled metric collection by placing the target database in a "blackout" status.
Metric Error	An error has occurred that is preventing metric collection. One common cause is incorrect EMCC agent credentials.
Pending/Unknown	Setup of the EMCC agent is in process but hasn't yet completed.
Target Down	The target database or instance is not running.
Unreachable	The "unreachable" message indicates a network problem, or the server has been disabled or turned off.

Stale Metrics

In addition to the statuses above, metrics will be missing when agents are not installed or if agents have been de-configured or otherwise disabled. In some cases, it's important to also evaluate the DATE of the last known status of a target. The last known status might be "Target Up" but with a very old date.

Metric Collection Disabled

Another source of missing metrics can be EMCC Monitoring Templates that exclude metrics from collection. If the customer has an active template that excludes metrics, those metrics are not available and cannot be extracted. The customer can modify the template and collect those metrics in the future, but substitutes will be needed to conduct a sizing exercise. Information about EMCC Monitoring Templates can be found in the EMCC Administrator's Guide section on Using Monitoring Templates. The EMCC extracts process generates the following file to help diagnose whether metric collection has been disabled:

- `emcc_sizing_em_metrics.csv`

Any metrics that have been disabled for a target will show ZERO ("0") in the IS_ENABLED column of this file.

Standby Databases

Standby databases are a special case of missing metrics as discussed in the next section. Standby databases don't report metrics to EMCC unless they are opened. If a Standby Database has been activated previously, those metrics will be in the repository but will be outdated. Data Guard Standby Databases are a special case of missing metrics that is outlined in more detail in a section to follow, including special tools in OCCA for handling these situations.

Errors During Extract Process

Another common cause of missing metrics is errors during script execution. Any errors during script execution will be reflected in the .csv or .txt files in the "extracts" directory. Search those files for "ORA-" error messages.

Correcting Missing Metric Issues

In some cases, the correct action is to exclude these databases from sizing, but it's important for the customer to weigh-in on the decision to exclude them. It is sometimes not feasible to correct all missing metric problems in a timely manner. It will take 30 days to collect 30 days of metrics, and 12 months to collect 12 months of metrics. For this reason, it will be necessary to fill gaps in data by entering the needed data in the following files:

- `properties_database.csv` *Always used*
- `properties_instance.csv` *Often used*
- `properties_server.csv` *Almost never used with OCCA*

Modifying information contained in the `emcc_sizing_extracts` directory is not supported. The supported method to inject data into the metrics is via these property files. Do not delete rows from any of the .csv files. Do not modify files other than the properties files listed above. Only `properties_database.csv` is required. Best practice is to omit the property files (by leaving the "original" string in the file names) if they aren't typically used.

Determining Values to Substitute for Missing Metrics

Only the customer can determine the proper value to use for substituting any missing metrics by simply querying each database. However, in the absence of customer input, one option is to simply use the AVERAGE across all databases within a cohort. For example, use the average database size for any databases that are missing size information. All substituted values should be included in a report to the customer.

Missing Instance Metrics

Databases will be excluded when 1 or more instances of that database have missing metrics. There are two basic approaches to handling missing metrics at the Oracle Database instance level as follows:

- Set Instance Metrics to Zero
- Copy Metrics from Another Instance

The correct answer for these often depends on the status of the instance at the time of data collection when the EMCC Extract process was run. The following table shows the most likely corrective action based on the state of the instance at time of extract. Customers might also give additional guidance based on their knowledge of their systems.

INSTANCE STATUS	SET TO ZERO	COPY FROM ANOTHER INSTANCE
Unreachable	X	
Target Down	X	
Pending/Unknown	X	
Blackout	X	
Metric Error		X
Target Up		X

Metrics can be copied from another instance that belongs to the same database by specifying the instance name in the "Use Values from Instance" column in the `properties_instance.csv` file. Metrics can be set to ZERO by specifying ZERO ("0") in the vCPU, SGA (MB), PGA (MB), IOPS, and Logons columns of `properties_instance.csv` file.

Understanding Data Guard Workloads

Oracle Data Guard can be used in several different implementations that affect whether metrics are reported to Enterprise Manager and the workload reflected in those metrics. In this section, we will cover the following implementations of Data Guard and how each affects sizing:

- Disaster Recovery Only
- Readable Disaster Recovery (ADG)
- Read-Only Replicas, Clones or Reader Farms (ADG)

The most common implementation of Data Guard is in a Disaster Recovery-Only mode, using standard (non-Active) Data Guard. Standard Data Guard standby databases do not report ANY metrics to Enterprise Manager unless they are activated.

Missing Metrics for Standby Databases

Metrics are often missing for standby databases due to a variety of reasons. It's not possible to determine the use-case of these standby databases simply through evaluation of metrics. Standby database metrics typically fall into one of the categories below:

- Missing ALL Metrics on Standby
- Not Missing Metrics
- Missing SOME Metrics on Standby
- Standby Missing Entirely

When all metrics for a registered standby database are missing, this might indicate a Pure Disaster Recovery Database, but might also be caused by an incorrectly configured EM Agent. In either case, the metrics are missing, and action must be taken to resolve this condition.

When a registered standby is NOT missing any metrics, this might indicate a Readable Standby Database, but does give metrics that can be used for sizing purposes. When a registered standby is missing SOME metrics, the use-case of the standby is not known. However, the partial metrics can be used for sizing along with some substitution values.

The last case is a standby database that is either missing from (not registered in) the Enterprise Manager repository used in the extract, or the associated primary database is not registered. Some customers run multiple EM Repositories, so primary and standby database pairs might be registered in different repositories.

Storage Sizing for Standby

Storage size for a Physical Standby database should always be copied from the associated Primary database. This eliminates issues that occur when a Standby database has not reported its size to Enterprise Manager and is a valid assumption because Physical Standby databases are physically the same size as the Primary aside from minor variations that occur over time.

Physical Compute Shape Sizing

The physical compute shape for a Standby database must be large enough to run the intended workload during a disaster.

- **Run Primary Workload** (Pure DR or Sacrifice of Read-Only Workload)
- **Run Primary Workload + Read-Only Workload** (Active Data Guard)
- **Run Read-Only Workload** (Reader Farm)

OCCA makes decisions based on the presence or absence of metrics, which may not be correct based on what the customer intends. It's not possible to discern intent through analysis of the EMCC metrics. Users may need to override the directives used by OCCA.

vCPU/OCPU for Normal Operation

Under normal operating conditions, Pure DR and Readable Standby databases will have lower processing demands than during a disaster. If metrics are NOT present, OCCA will traverse DR relationships to find the associated primary database and use 20% of the primary vCPU metrics by default (the 20% value can be modified). If any metrics ARE present (not necessarily all metrics), OCCA will attempt to use those metrics for vCPU/OCPU sizing. User intervention will be required in cases of partial metrics.

vCPU/OCPU for Disaster Operation

OCCA will automatically attempt to size for Disaster operation based on presence or absence of metrics. If metrics are missing, OCCA will size Standby for disaster condition using metrics of the primary. If metrics are at least partially present for the standby, OCCA will size Standby for running BOTH the primary workload as well as the read-only (standby) workload. Some customers find it impractical to sacrifice the read-only workload during a disaster, so this is the default behavior. If a customer intends to sacrifice (turn off) the read-only workload during a disaster, the user will need to override the directive in `properties_database.csv`.

OCCA Controls for Copy DB Metrics

OCCA provides a number of controls for the handling of Data Guard Standby databases. These controls are set automatically by the Copy DB Name command (`occa --copy-db-name`) but can be overridden using the following columns of `properties_database.csv`.

Column Name	Contents
Use Values from Database Name	Specifies the Name of a Database to Copy Values From. Name specified must be in the same EMCC Extract.
Use Values Scope	Directives for handling Copy (valid values) 1. Physical Standby 2. Primary + Standby Instance Metrics 3. Standby Instance Metrics 4. All
Use Values Pct	Percentage to use when copying Metrics (positive numbers)

These values are set by OCCA (`occa --copy-db-name`) and are based on the presence of metrics, which may not be what the customer intends. Details of how these settings impact sizing are outlined in the following sections.

Directive: Physical Standby

The directive "Physical Standby" is set automatically when Standby metrics are missing. This directive sizes the standby at 20% of the primary during normal operation, and 100% of the primary during a disaster. The 20% value can be overridden as outlined previously.

Standby Sizing Attribute	Normal Operation (Standby Mode)	Activated During Disaster
Database Storage	100% of Primary	100% of Primary
Instance Memory	100% of Primary	100% of Primary
vCPU Consumption	N% of Primary (default 20%)	100% of Primary
I/O Consumption	N% of Primary (default 20%)	100% of Primary
Sessions	100% of Primary	100% of Primary

Directive: Primary + Standby Instance Metrics

The directive "Primary + Standby Instance Metrics" is set automatically when at least SOME Standby metrics are present. This directive assumes that the standby will need to run both the primary workload as well as any read-only workload that normally runs on the standby. If the customer intends to sacrifice the read-only workload during a disaster, the directive can be changed to "Physical Standby".

Standby Sizing Attribute	Normal Operation (Standby Mode)	Activated During Disaster
Database Storage	100% of Primary	100% of Primary
Instance Memory	100% of Primary SGA/PGA + 100% Standby PGA	100% of Primary SGA/PGA + 100% Standby PGA
vCPU Consumption	100% Standby	100% of Primary + 100% Standby
I/O Consumption	100% Standby	100% of Primary + 100% Standby
Sessions	100% of Primary + 100% Standby	100% of Primary + 100% Standby

Directive: Standby Instance Metrics

The directive "Standby Instance Metrics" specifies that the standby will always run the read-only workload and log apply workload and will never be used to run the primary workload during a disaster. This directive is never used automatically by OCCA and always needs to be set manually because it's not possible to determine customer intent by simply analyzing data.

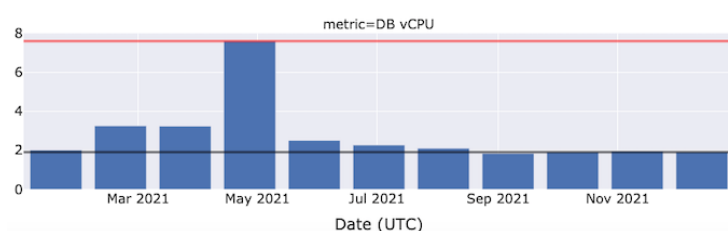
Standby Sizing Attribute	Normal Operation
Database Storage	100% of Primary
Instance Memory	100% Standby Metrics
vCPU Consumption	100% Standby Metrics
I/O Consumption	100% Standby Metrics
Sessions	100% Standby Metrics

Directive: All

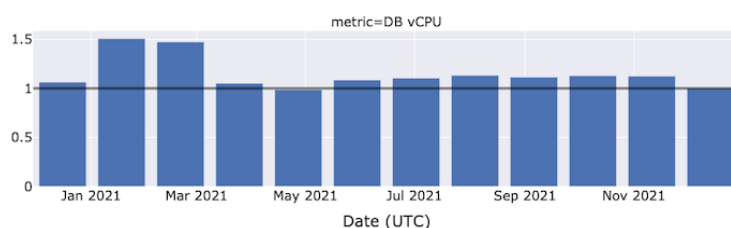
The Directive "All" allows the user to copy ALL metrics from one database to another. It is intended to be used for cases where database or instance metrics are missing. This directive is never used automatically by OCCA.

Adjustment Plots for DR Workloads

Disaster Recovery (DR) workloads will often show periods of high activity during the year, especially when customers conduct periodic DR tests. These peaks are often more dramatic in cases where DR databases are arranged into dedicated DR cohorts as seen in the following example produced WITHOUT the "Use Values from Database" option:



While it is not possible to know what occurred by simply looking at the data, the scenario above is most likely due to a Disaster Recovery test conducted in the April/May timeframe. The "Use Values from Database" option will follow the same ratios in adjustment plots from the primary as shown below:

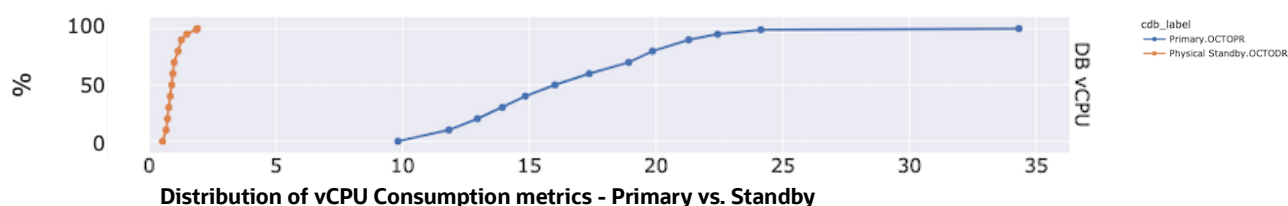


DR Adjustments Multiplier Plot with "Use Values from Database"

The MULTIPLIER version of this plot shows the multipliers used for OCPU calculations, matching the same multipliers used for the associated Primary database cohort. The plot of vCPU is used for physical sizing and shows the same size as the Primary database cohort. The "Use Values from Database" option intentionally suppresses prior peaks that might have occurred in Standby Database processing.

Reduce DB Metrics

Additional analysis can be performed on Primary vs. Standby relationships for cases where all necessary data is available. The volume of DB metrics is too large for analysis with tools such as Excel, so OCCA provides the ability to reduce the volume of data. OCCA also provides some detailed charts that facilitate the analysis of Primary vs. Standby resource to help validate the default 20% consumption estimate. The command "**occa --reduce-db-metrics**" produces charts for each resource consumed by each database. The following chart shows the distribution of vCPU consumption of a primary database (in blue) and the associated Standby (DR) database (orange line).



The above chart shows that the Primary database is using nearly 35 vCPU, while the associated Standby database is using a maximum of approximately 3 vCPU. This data shows that the default of 20% is quite conservative because the Standby is consuming approximately 10% of the vCPU resources that the Primary consumes. For any cases of missing metrics, those metrics simply will not be present in these charts.

Sizing, Capacity Planning, Consumption, and Performance

Sizing and performance are related, but ultimately separate topics. Under-sized systems can impede performance, while over-sized systems will not improve performance. Anytime demand reaches the limit of supply, the true level of demand cannot be measured. Consumption of resources is demand, whereas supply is the number of resources available on a system.

Sizing

The 4 primary dimensions of system sizing are CPU, memory, I/O and storage space. Exadata is a "balanced" system, which allows us to focus on these 4 primary dimensions. Unbalanced systems may suffer slow response time caused by other components such as storage network, cluster interconnect, etc. Consequently, the methodology used for sizing Exadata doesn't necessarily work well for sizing non-Exadata systems.

Resource Consumption

Consumption refers to how much resources a database consumes, including CPU, Memory, I/O and storage space. Consumption should be measured over a period of time in order to characterize daily, weekly, monthly, and even annual peak processing periods.

Capacity Planning

Sizing should be done based on the current system resources and consumption of those resources. Future growth (capacity planning) needs to be addressed AFTER sizing to replace existing systems on an equal basis. Customers often over-estimate future growth needs, so we need to be clear about sizing for the current footprint vs. how their estimated growth will be absorbed.

Performance Tuning

Performance tuning can change the consumption of resources measured on a given set of databases and therefore will change sizing requirements. However, we recommend users perform sizing independently from performance tuning. A sizing exercise will give more accurate results if performance is being addressed independently. Attempting to address performance tuning during the sizing process would result in a never-ending cycle of tuning and sizing.

Understanding Exadata Performance Benefits

Sizing does not consider the performance benefits of Exadata and does not account for increases in performance from one generation of Exadata to the next. Exadata is widely known to deliver BETTER performance than competing systems, including traditional on-premises architectures as well as Cloud Infrastructure from Amazon (AWS), Microsoft Azure, and Google Cloud Platform (GCP). The primary gains in performance on Exadata come from the following when comparing against non-Exadata and on upgrade from older generations of Exadata:

Exadata New Capabilities	Versus Non-Exadata	Upgrade from older Exadata
PMEM & RDMA/RoCE - I/O Latency	✓	✓
RDMA/RoCE - Interconnect Latency	✓	✓
Faster PCIe Bus		✓
Larger Storage Capacity		✓
KVM - Improved Virtualization		✓
Storage Offload	✓	

The result of these Exadata innovations might result in lower vCPU consumption compared to platforms from other on-premises or Cloud vendors, but no adjustments are made in the sizing process because of these innovations. The OCCA sizing methodology is designed to be conservative by avoiding under-sizing.

Percentile Calculations

OCCA calculates resource consumption percentiles and displays the resulting values in the OCCA Analytics plots. The following percentile calculations are displayed as **blue dotted lines** in each plot:

- 95th Percentile
- 90th Percentile
- 80th Percentile

OCCA calculates the **PEAK** resource consumption (**red line**) observed within the past ~33 days of detail data, adjusted to any prior annual peak if the annual peak is not within the past ~33 days. OCCA also calculates the **FLOOR** of utilization on an hourly basis (black line) as well as the average of the hourly floor (**green line**). Percentiles are provided to aide discussion of the potential savings that could be realized by adopting a more active OCPU management plan (see the "OCPU Management Plan" section of this document for details).

Sizing Data Collection

Data can be collected in several ways as outlined below. The accuracy of sizing is critically dependent on the accuracy of data used in the sizing effort, so we recommend careful attention to data collection.

OCCA AWR Miner

OCCA AWR Miner gives a better view of resource consumption for a single database than standard AWR reports, which only show highly detailed metrics for a single time interval. OCCA AWR Miner gives the full TREND of key sizing metrics available in the AWR History tables, with a typical retention period of 8-30 days. AWR Miner was designed to report on specific database performance metrics that highlight database performance metrics that Exadata is specifically designed to address but does not give sufficient time-period to evaluate seasonality over the course of a year due to lack of data in AWR History.

EMCC Extract Scripts (Inventory & Resource Consumption)

EMCC gives a more complete view of inventory and much longer history of resource consumption. EMCC cannot automatically distinguish data center locations and cannot automatically determine which databases need to remain separate from each other, so we cannot automatically determine the proper cohort for databases.

EMCC Extract & AWR Miner Licensing Exception

The EMCC Extract Scripts and AWR Miner do not require special licensing as outlined here: OEM Metrics License Exception for Analysis by Oracle Pre-Sales or Consulting (Doc ID 2771065.1).

Manual Input

Sizing requires customer input regarding how databases should be arranged into cohorts, as well as to fill any gaps that might arise in the data collection process. EMCC systems frequently include incomplete information as addressed in "Step Zero" (EMCC Health Assessment) later in this document.

EMCC Extract Scripts

The EMCC Extract scripts are designed to extract data from a single EMCC repository in order to generate sizing for all databases managed through that repository. Please see the REFERENCES section of this document for download instructions.

Versions Supported

The EMCC data collection process works with the following EMCC Versions:

- EMCC 13.x and later

Version of EM Agents installed on each target is EXTREMELY IMPORTANT because this impacts the data collected by the agent for that target. Older agent versions do not collect important metrics used by OCCA and may require overrides (entered in Properties files).

SQL Plus Version Requirement

SQL Plus version 12.2 or later is required for executing the scripts written for this process. Earlier versions of SQL Plus will result in errors during processing. The version of SQL Plus is reported in the sizing tool output, and we give guidance below to help identify if an older version of SQL Plus was used. Repair scripts are available if the customer uses an older version and is unable to use the correct version.

EMCC & Pack License Requirements

The EMCC extract and OCCA AWR Miner scripts do not require Diagnostic & Tuning licenses as outlined in MOS Note# 2771065.1 (OEM Metrics License Exception for Analysis by Oracle Pre-Sales or Consulting).

Oracle Employee Laptop Setup

The following includes directions for setup of Oracle employee laptops. This is required for running the Python scripts and shell scripts (or PowerShell files for MS Windows). Users of MS Windows need to adapt these instructions to their environment.

1) Install Laptop Tools

- a. Install standard Python3 binaries for the Desktop platform
 - i. <https://www.python.org/downloads>
 - ii. Choose a relatively current version but stay away from versions less than a couple of months old.
 - iii. Windows: Check the "Add to Path Option" to add Python to your path
- b. Connect to VPN and navigate to the OCCA Web Page, and Learn Sizing
 - i. <https://occa.us.oracle.com/ords/r/occa/occa/occa-releases>
 - ii. Locate the latest OCCA (Python) Desktop Analytics wheel file and download
 - iii. File format will look something like the following:

occa-<YY>.<M>.<SEQ>-py3-none-any.whl

- <YY> Is the 2-digit-Year
- <M> Is the Month
- <SEQ> Is optional sequence number

Example: occa-26.3.0-py3-none-any.whl

- c. Disconnect from VPN and run:


```
pip3 install occa-26.3.0-py3-none-any.whl
```
- d. Ensure "occa" executable is in your path:

MacOS

Location: /Users/<user>/Library/Python/3.11/bin
 Modify Path: export PATH=\$PATH:/Users/<user>/Library/Python/3.13/bin
 Shell Profile: .zprofile

Windows

Location: C:\Users\<user>\AppData\Local\Programs\Python\Python313\Scripts
 Windows Path: Execute the command "PATH" to check this setting

Keep OCCA Package Updated

The Python code on your desktop needs to be kept up to date. Be sure to download the latest version before each sizing engagement. If you are connected to VPN, the "occa" desktop command will automatically search for the latest version and alert you if a newer version is available.

OCCA for EMCC Extracts - Directory Structure

The OCCA Analytics code on the laptop requires the extracted .CSV files to be located in a specifically named directory, but beyond that the other directories listed below will be created by the application.

in the uses set of directories created on your computer as outlined in the table below.

Directory	Description
<work_dir>/	Top level directory where the extracted files are unzipped and where all the output files will be created. Each EMCC Extract will require its own Work_dir
emcc_sizing_extracts/	Directory within the Work_dir where the extracted CSV files must reside. The user must create this directory and unzip the files into this location.
occa_sizing_properties/	OCCA generated Property files used for creating Cohorts and handling missing values
occa_sizing_output/	OCCA generated sizing file, can be viewed to identify missing values but cannot be updated by user
duplicate_tests/	
log/	Log files from script execution

override/	
plots/	Plots generated from OCCA sizing output files
<plot_type>	Either <i>adjusted</i> , <i>DailyUnadjusted</i> , or <i>unadjusted</i> (optional)
<cohort_name>/	Directory for each cohort
html/	HTML versions of resource consumption plots for the cohort
png/	PNG image versions of resource consumption plots for the cohort
simulator/	
sizing/	Key Sizing outputs used to identify missing data, and ultimately the source of the data uploaded to the OCCA Web Application

Customer Scripts

Scripts for the customer to run are contained in a separate ZIP file that can be sent to the customer. Be sure to download the latest version before each sizing engagement. The current version of the code is always posted on the Oracle network (see appendix).

Source Side Filtering

The tool currently does not include the ability to directly filter data from Enterprise Manager at the source however the customer does have the option to use a tool provided to exclude databases from the results that are sent back to Oracle.

It is sometimes possible to manually modify SQL in the extract scripts, but this often renders the output unusable if not done correctly. In **extreme situations** if this becomes required, please reach out to the OCCA team on the Slack Support Channel.

Source Side Obfuscation

Much like the Source Side Filtering the tool does support the option to obfuscate the data sent by to Oracle. If this is required the customer can use the tool provided to obfuscate the names within the extracts and the key is left with the customer.

Obfuscation adds a layer of complexity to the processing and understanding of the data that the customer provides so this should only be used when required by the customer's security policies.

to manually modify SQL in the extract scripts, but this often renders the output unusable if not done correctly. In **extreme situations** if this becomes required, please reach out to the OCCA team on the Slack Support Channel.

Tool Demo Data

Demo extract data for Oracle employees to learn how to size systems without needing to access customer data is available on the Oracle network (see appendix).

OCCA Sizing Process

The OCCA sizing process allows us to collect much more accurate information, producing more accurate sizing than ever before. All comparative sizing efforts require the following inputs:

- Inventory databases and servers they reside on
- Database consumption of resources

The OCCA sizing process allows us to execute a SINGLE extract that captures both the database inventory as well as resource consumption for hundreds or even thousands of databases, servers, and virtual machines.

Process Overview

The following provides an overview of the process that is outlined in detail in the sections that follow.

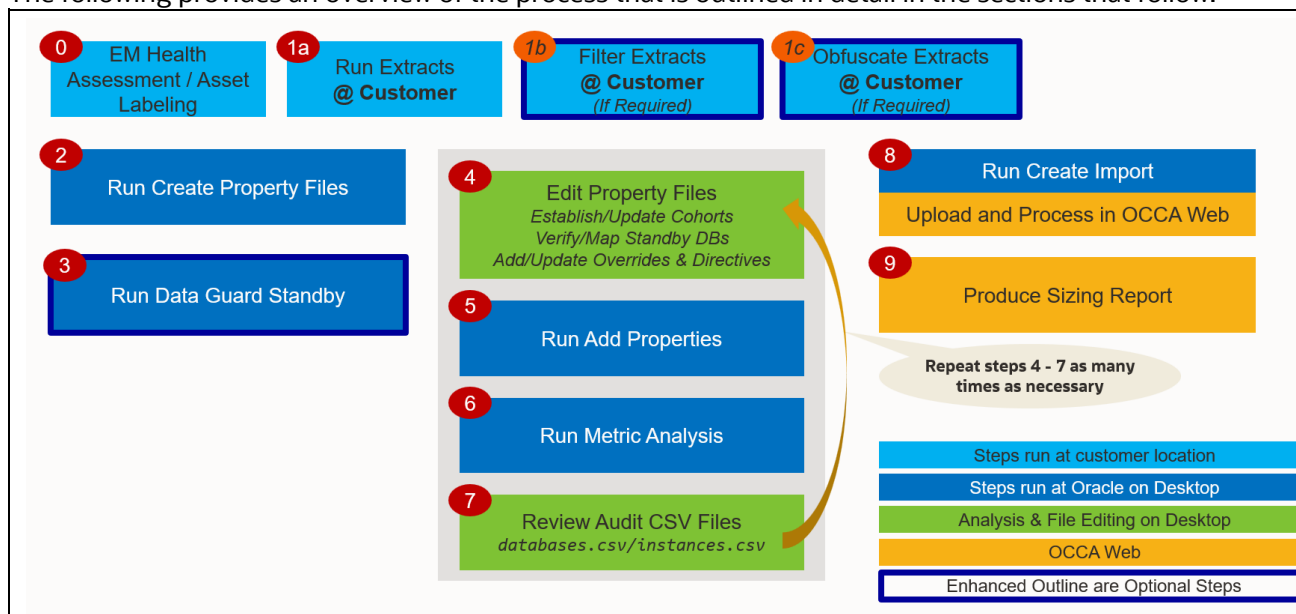


Figure 1: OCCA Sizing Process Overview

Step #0: EMCC Health Assessment

Prior to an engagement, it's important to have a discussion with the customer regarding the health of their EMCC system. A healthy EMCC system should not have issues when running the EMCC extract scripts used in this process.

- Beware of potential routine care & maintenance issues
- EMCC often contains retired servers and databases
- Targets may have agents that are down and not sending data
- Asset Labeling might not be used
- Available history depends on age of EMCC deployment
- Available history depends on age of targets

While we are not providing guidance currently on how to conduct an EMCC health assessment, it's important to be aware that the relative health of the EMCC system can be an issue during the data collection process and may result in missing data that will need to be provided manually during this process.

Pre-Flight Check

OCCA Analytics includes a command to generate a simple report to help judge status of Enterprise Manager targets. The command **occa --pre-flight** can be used to generate the following report:

Target Type	Blackout	Metric Error	Pending/Unknown	Target Down	Target Up	Unreachable
Host	23	0	0	0	500	6
Oracle Database	57	0	1	6	1287	6
Oracle PDB	39	2	1	14	680	8
RAC Database	31	2	2	3	813	0

This command is based on extracted data and needs to be run after step #1a, 1b or 1c. The pre-flight Python script can be sent to the customer so they can run it, or the script can be run by an Oracle employee using the command above. The command generates 2 files as follows, including the summary above as well as a detailed listing of targets and statuses.

- sizing/current_status_by_target_type.csv
- sizing/current_status_by_target.csv

Step #1a: Customer Executes EMCC Extracts

The customer needs to execute an EMCC extract script against their EMCC Repository. Customers might have a single EMCC repository, or multiple repositories. Some customers use separate EMCC systems for DEV/TEST as opposed to PROD. Other customers might use a separate EMCC system for each data center, while others might use an EMCC system with a Data Guard copy of the database for DR purposes. Be sure you understand how the customer uses EMCC and what are the contents you will receive.

- 1) Send/Share scripts to customer (Customer_emcc_sizing_extracts.zip)
 - Oracle Email rules will not allow this file to be emailed, use tool such as SharePoint
 - The customer will need to return/upload results again using a tool such as SharePoint
- 2) Assist customer with executing scripts
 - The zip file has the EMCC-Extracts-Guide.pdf file with detailed instructions for running the Extracts.
 - On Linux/Unix run: `Get_EMCC_Interactive.sh` and it will prompt for required inputs
 - On Windows run: `Get_EMCC_Interactive.ps1` and it will prompt for required inputs

Step #1b Optional: Source Side Filter

- a) For customers who cannot share all of their extracted data
- b) Filtered data can make it more difficult to drill into and resolve issues
- c) Customer will need to install Python (3.x or later)
- d) Refer to the EMCC Extracts User's Guide included in the zip file for detailed steps
 - Customer runs: `python Filter_Extracts.py --create-filter` to create the CSV file
 - Then updates the file, `properties_database_filter.csv`, to identify DB's to exclude
 - And then runs: `python3 Filter_Extracts.py --filter`, to create the filtered output
- e) Results from this step are in directory: **emcc_sizing_extracts_filtered**

Step #1c Optional: Source Side Obfuscation

- a) For customers who cannot share server names, database names, owners, etc.
 - b) Obfuscated data can make it more difficult to drill into and resolve issues
 - c) Needed especially for National Security customers
 - d) Customer will need to install Python (3.x or later)
 - e) Refer to the EMCC Extracts User's Guide included in the zip file for detailed steps
 - Customer runs: `python Obfuscate_Extracts.py`
 - Customer retains the Extracts Translation File (`obfuscated_translation.csv`)
 - f) Results from this step are in directory: **emcc_sizing_extracts_obfuscated**
- 3) Customer Zips up the results output directory and uploads to shared folder provided by Oracle
 - Default Directory: **emcc_sizing_extracts**
 - Filtered Directory: **emcc_sizing_extracts_filtered**
 - Obfuscated Directory: **emcc_sizing_extracts_obfuscated**
 - 4) Receive extracts from customer
 - a. On your desktop, create a customer-specific working directory, e.g. : `<cust_name/work_dir>`
 - b. Rename customer-provide Zip file, if necessary, to "emcc_sizing_extracts"
 - c. Unzip into `<rootdir>/<cust_name/work_dir>/emcc_sizing_extracts`
 - Ensure extracted .csv files are in the "emcc_sizing_extracts" directory
 - 5) Review Output Files
 - a. Ensure no ERRORS in output files (ex: ORA-xxxxx errors)
 - b. Review files for overall correctness
 - Check version of SQL*Plus used is 12.2 or above (interactive script prevents this)
 - CSV files should look correct
 - TXT files should only contain start and stop times of each step

Step #2: Create Property Files

The "properties" file allows us to provide additional information that is not contained in the automated extract. The properties files are generated automatically by running a script.

- 1) Run the Script: **occa --create-properties**
- 2) This generates empty files that are used in the next step
- 3) Copy the "original" properties files, removing the "_original" from the names. Don't modify the "original" properties files, for example:

```
copy occa_sizing_properties\properties_database_original.csv
      occa_sizing_properties\properties_database.csv
copy occa_sizing_properties\properties_instance_original.csv
      occa_sizing_properties\properties_instance.csv
```

Step #3: Data Guard Metrics

Data Guard Standby (pure DR) Databases do not report metrics to EMCC unless they are opened. DG Standby Databases that are never opened will never report metrics. Standby Database metrics will often be outdated if the database had previously reported its metrics. To address this condition, the command below copies Data Guard Standby Database metrics from the associated primary database.

Run the command: **occa --copy-db-name**

This automated process handles the case where Standby and Primary Databases are managed within a single EMCC Repository by traversing the relationship of each Physical Standby to its associated Primary Database. Default settings are as follows:

- **Use Values Scope Default:** Physical Standby
- **Use Values Pct Default:** 20 (20%)

If the "Use Values Scope" and "Use Values Pct" columns are left blank, the default during processing (in Run Metrics Analysis) is to copy vCPU and IOPS metrics from the associated Primary Database at 20%, while Memory and Storage will be copied at 100%.

Active Data Guard Standby Databases will report metrics because they have been opened. ADG databases are often used for running a QUERY workload and sustain the REDO apply workload. Please refer to the section of this document on "Understanding Data Guard Workloads" for more information.

Note: The steps above are one and done, you don't need to rerun them after any of the following steps, unless you want to start over, in which case it is recommended to remove or rename the directories created by the OCCA tool.

The next several steps may need to be run multiple times, the basic process from here is setup cohorts run the processing and check for any missing metrics, add override values or set directives in the database and/or instance property files, run the processing and check again. Between addressing missing data and mapping the databases to the cohorts it is typical they you have to do this at least 2-4 times.

Step #4: Edit Properties

Extracted data should not be altered directly. If data is missing or invalid you must use the below property files to correct the issue. This is accomplished by adding override values, setting a directive, or explicitly excluding the database.

- 1) For each DB in the database property file, **properties_database.csv**, you must provide a value for column A (Cohort), options include:
 - Cohort Name
See the section of this document that explains the definition of "cohorts"
 - Keyword: **"excluded"** this is case specific
Excluding a database automatically excludes all mapped instances

- Blank
Database will be marked as unassigned and will be skipped and not uploaded

How to Exclude Databases

Databases often need to be excluded from sizing. There are 2 options for excluding databases as follows:

- Use keyword "**excluded**" for the cohort's name
- Leave "cohort" blank (defaults to "unassigned" and will be skipped)

Copying Properties from EMCC Extracts

If you have previously sized this opportunity using the older EMCC Extracts version, you can copy properties from the old output files using this command:

Run the command: **occa --copy-sizing**

Step #5: Add Properties

In this step, values from the properties file are added to the data that was extracted by the customer.

- Run the command: **occa --add-properties**

Step #6: Run Metric Analysis

This step runs analysis of the metrics contained in the extract files, grouping databases into cohorts as indicated, and creating the plots that will be used to show resource consumption for each cohort as well as for databases individually in isolation.

- Run the command: **occa --run-metric-analysis**

Step #7: Review Audit Information

1) Review audit information on Databases and Instances

- Directory: `occa_sizing_output/sizing`
- Files:
 - `databases.csv`
 - `instances.csv`
 - `servers.csv` *(informational only)*
- Look first in `databases.csv`, if the DB was implicitly excluded due to missing metrics it will be identified in column A with the reason in column B.
- Common issues that cause databases to be excluded include the following
 - Database is missing required fields
 - Instance is missing required fields
- The root cause for missing values can be hard to pin down.
 - It could be that the agent is down, or there is a metric collection error, check the "status" and the last load time columns.
 - It could be that the monitoring of the particular metrics was disabled in EM via a monitoring template.
 - An instance will typically fail if instance metrics (vCPU, Memory, and I/O) are missing in the extract. A database will fail if the DB size allocated or used values are missing.
 - For the current sizing effort, the why it's missing typically does not matter, if the database is to be included in the sizing a value will be needed.
- Missing values can be provided manually, or all metrics can be automatically copied from another database in the extract files using a directive such as the "Use Values from Database Name" setting in the `properties_database.csv` file.
- Standby Databases should have metrics copied from the associated primary. See step #3 of this document for more information on handling of Data Guard Standby Databases. See the

section of this document Handling Missing Metrics for more information about why metrics can be missing and what actions should be taken to address.

2) Sanity Check on Extracts

- a. Ensure customer has sent all extract files (both **.csv** and **.txt** files)

Note: If the customer performed Source Side Filtering or Source Side Obfuscation, then the number of CSV files may be fewer than expected, and the TXT files will probably be missing.

- b. Look for Oracle Errors in Extract Files

- In the extracts directory: `cd <cust_name/work_dir>/emcc_sizing_extracts`
- Run a command like this: `grep "ORA-" *`

Severe errors in processing will occur if the customer has modified the extracted data or modified the scripts that collect the data. Oracle Database (ORA- errors) in the extracts will also result in severe errors during processing. Customers need to provide ALL extract information as outlined in step #1 of this document, including both DAILY and HOURLY metrics.

Note: If databases have been implicitly excluded, or something does not check out in the sanity check, the next step is to go back to step #4. If you attempt to run the import command with errors and implicitly excluded databases it will fail.

Step #8: Upload and Process Opportunity in OCCA

- 1) Run the command: `occa --emcc-import`
- 2) Login to OCCA: <https://occa.us.oracle.com>
- 3) Navigate to "Upload Extracts" and upload the file
 - a. Upload this file: `<cust_name/work_dir>/occa_sizing_output/occa_upload_data.json`
- 4) Review Uploaded Data for accuracy/completeness
 - a. List of Databases (cross-check against: `<work_dir>/occa_sizing_output/sizing/databases.csv`)
 - b. Cohort Rollups (cross-check against: `<work_dir>/occa_sizing_output/sizing/cohort_rollups.csv`)
- 5) Set Desired Target(s) in OCCA
 - a. Choose selected target platform(s) and options
 - b. Choose the same target for all cohorts or different targets for each cohort
- 6) Run Sizing Report & Review Results
 - a. Review Exadata and/or BaseDB sizing results
 - b. Review Sizing Output Messages (debugging tool)
 - c. Review Sizing Overview

Step #9: Produce Sizing Report

The sizing report provided to the customer should include a number of elements to ensure the customer understands how the sizing was calculated. The following section of this document includes a discussion of what elements should be included in the report.

Flash Cache Sizing

Exadata systems cache data into Flash Cache and XRMEM (Exadata RDMA Memory) cache to speed processing while also delivering the low-cost benefits of HDD storage. This applies to all High-Capacity configurations of Exadata across all deployment models including Exadata on-premises, Exadata Cloud@Customer, Exadata in OCI, and Oracle Autonomous Database. Exadata X10M Extreme Flash also uses a Flash Cache layer on Performance Optimized Flash, while permanent storage resides on Capacity Optimized Flash. There are 3 options in OCCA for sizing Flash Cache as follows:

OPTION	OCCA WEB SELECTION
Size Disk (or Capacity Flash) for Allocated Storage	Force Used in Flash = NO

Allocated storage is the total of all Tablespace Sizes.	
Size Flash Cache for USED Storage USED storage includes all of the space that is used within the allocated tablespaces. The amount of USED space will be less than what is allocated.	Force Used in Flash = YES
Size Flash Cache as Percentage of ALLOCATED Storage Allocated storage includes all the space allocated to the tablespaces. Allows selection of the percentage.	Force Used in Flash = PCT
Size Flash Cache based on Working Set Size The Working Set of any database is the amount of data that is actively being used (read or written) within the USED space.	Force Used in Flash = AUTO

OCCA AWR Miner captures information about the Working Set Size by evaluating the I/O characteristics posted to database segments. Segments in an Oracle Database include tables, partitions, and indexes. Oracle Database collects I/O statistics at the segment level using the `gv$segment_statistics` dynamic performance view.

DB Segment I/O Analysis Limitations

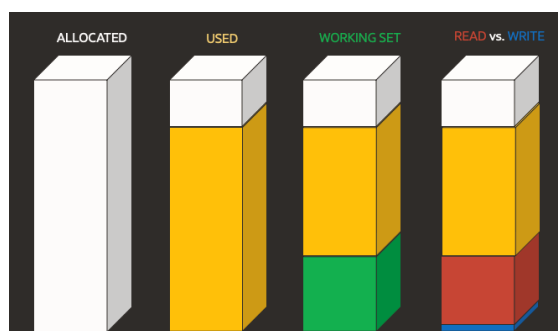
There are several important limitations that affect the analysis of DB Segment I/O information in Oracle Database as follows:

- Only the non-CDB database architecture is supported, Multitenant support is not available
- Segments get created and dropped
- Library Cache Space Pressure
- Contents are cleared on Instance Restart

Segment statistics are tracked at the segment level, so frequent creation and drop of segments will affect the statistics available for analysis. Space pressure in the database Library Cache can affect the contents of `gv$segment_statistics`, so the information may not be complete. It's also important to understand that an instance restart will clear the contents of all Dynamic Performance Views, so the contents are only valid for the time period since the last instance restart.

Allocated vs. Used vs. Working Set

The following graphic illustrates the differences between the amount of storage allocated to a database, the amount of storage used within the database, and the working set size.



For example, a database might have 100TB allocated space, 80TB used within that space, and only 33 TB active or what is known as the Working Set Size. In addition, the Working Set might consist of a workload that involves 90% read (30TB) and 10% write (3TB). Exadata caches blocks subject to READ in 2 Storage Servers, while blocks are written to 3 Storage Servers using High Redundancy.

Importance of Flash Cache Sizing

It is extremely important to ensure that Flash Cache is sized properly for a given workload. Flash offers much higher I/O performance than HDD. If the Working Set Size exceeds the Flash Cache size, the I/O capabilities of HDD might not be sufficient for the workload.

Recommendations for Sizing Reports

The final report is prepared using a combination of information from OCCA Analytics and OCCA Web. This section outlines recommended content that should be provided in reports to customers.

Recommended Contents

All sizing reports should contain a number of standard elements so that customers understand how sizing was performed and what was included.

- Standard Disclaimer
- Summary of Missing Metrics & Steps taken to compensate
- Summary of Cohorts, number of databases, etc. (see Cohort Rollups)
- Summary of Cohorts and Sizing for Each (see Sizing Report)
- Plot of Adjusted vCPU by Cohort (occa_sizing_output/plots)
- Plot of Adjustments & Seasonality by Cohort (occa_sizing_output/plots)
- Screenshot of OCCA Web chart showing vCPU required vs. configured

In addition, sizing for Cloud targets that support consolidation should also include estimates of OCPU charges, which are variable and managed by the customer.

Summary of Missing Metrics

The summary of missing metrics should begin with a listing of database statuses and count of databases with each status. The following report is created using a Pivot Table in Excel with data from "properties_database.csv" with ROWS = cdb_status and VALUES = count of Database Name:

DB Status	Count of Database Name
Blackout	52
Metric Error	2
Pending/Unknown	3
Target Down	6
Target Up	1171
Unreachable	4
Grand Total	1238

As we see from the above table, this customer has 1,238 databases in total registered in Enterprise Manager, with the vast majority (1,171) in "Target Up" status. As noted earlier in this document, Standby Databases are a special case of missing metrics and OCCA provides tools for addressing these. The following Pivot Table shows a summary of standby databases registered in the Enterprise Manager Repository.

Standby Type	Count of Database Name
N/A	811
Physical Standby	214
Primary	213
Grand Total	1238

Note that the number of Physical Standby databases is different than the number of Primary Databases, which indicates an issue that might need to be addressed. In this example, there is 1 Physical Standby that does not have an associated Primary database in the EM Repository. The Primary Database might be

registered in a different EM Repository, or the Primary Database might not be registered at all. Note that "Standby Type" will be blank (N/A above) if a database is neither a standby, nor primary.

Similar Pivot Tables can be created using "properties_instance.csv" if there are widespread problems with missing instance metrics. OCCA does not use server/host metrics, so any missing server/host metrics will not impact sizing and we don't anticipate that any reporting of missing server metrics will be required.

Additional drill-down into missing metrics might be necessary depending on the number of affected databases or based the sizes or criticality of the affected databases.

OCCA Reports

OCCA generates reports for Exadata targets (Exadata on-prem, ExaDB-C@C, and ExaDB-D) in one report, and a separate report for any BaseDB targets. Each report includes "compact" and "full" report formats. The "compact" format should be appropriate for most circumstances. These are APEX Interactive Reports, which include some standard functionality that will be familiar to any APEX users or developers.

Customizing APEX Interactive Reports

APEX Interactive Reports can be customized by each user, and users can save their own report customizations as well.

Downloading Reports

Use the Actions→Download menu on the report to download the report in .csv format. The resulting file can be opened using Excel and copied into a table in PowerPoint.

OCCPU Sizing (Cloud)

The physical systems are sized based on the MAXIMUM calculated OCCPU consumption expected for each environment. For ExaDB-C@C and ExaDB-D deployments, the customer will then need to understand how to handle OCCPU scale-up and scale-down. The following plots generated by the EMCC sizing tool show the resource consumption graph over the past 12 months. These graphs are used to drive a conversation with the customer about how they would like to manage their OCCPU settings.

Note: OCCPU for Operating System and Grid Infrastructure

OCCA performs sizing based on consumption of resources for each database, which does not include any resources consumed by the Operating System (O/S) and Oracle Grid Infrastructure (Oracle Clusterware and ASM). The final calculations for OCCPU include the addition of 4 vCPU (2 OCCPU) for each Exadata Database Server to account for O/S and GI resource consumption.

For BaseDB High Availability (HA) configurations, OCCA adds 2 vCPU (1 OCCPU) to the account for O/S and GI, whereas non-HA configurations have 1 vCPU (1/2 OCCPU) added to account for O/S only. BaseDB systems are smaller and do not support database consolidation, so the O/S and GI are only servicing a single database or Container Database using Oracle Multitenant.

OCCA performs analysis in terms of vCPU, then makes the transition to OCCPU after the target is selected and additional CPU consumption for O/S and GI have been added. The "Minimal" report format shows the final sizing calculated by OCCA in the example below:

Exadata Sizing Report Output								
Q Search: All Text Columns Go Minimal Actions Reset								
Cohorts	Metrics			Resulting Config				
Cohort	Allocated Storage (GB)	DB Servers Required	Storage Servers Required	DB Servers	OCCPU Peak with OS+GI	OCCPU Peak for Standby with OS+GI	Storage Servers	Total Storage (GB)
AWS-US-EAST-2	18,582	0.31	0.45	2	17.00	16.00	3	153,600
AWS-US-EAST-1	166,940	1.49	4.08	2	64.00	62.00	4	208,674
EDC-North	73,149	0.98	1.79	2	26.00	19.00	3	153,600
Unknown	840	0.04	0.02	2	5.00	5.00	3	153,600

Exadata Cloud systems (Cloud@Customer and Cloud Service) are physically sized to handle the maximum expected workload, and OCCA generates OCPU sizing for databases running in "normal" (standby) mode as well as under peak processing loads expected during a disaster.

OCPU Peak for Standby with OS+GI

Estimated OCPU charges in a "set and forget" management model (the RED line from the vCPU Consumption plots) are based on the value for "OCPU Peak for Standby with OS+GI" for each cohort. This OCPU setting is for the case where any standby databases within the cohort are running in normal standby (log apply) mode.

OCPU Peak with OS+GI

This value is used to drive the physical sizing of the configuration required for each cohort, covering the case where ALL standby databases within the cohort are activated such as in a Disaster Recovery situation. The system needs to be physically sized large enough to accommodate this level of workload. However, the customer would scale down the primary site during a disaster, so OCPU estimates should NOT be based on this number.

Combining Cohorts Using VM Clusters

The defined cohorts might be much smaller than an Exadata system can support, resulting in over-sizing and higher costs than expected. In these cases, cohorts can be combined together onto an Exadata system using Virtual Machine Clusters. The value for "DB Servers Required" shows the number of Database Servers required by the workload within the cohort. Multiple rows can be combined together onto VM Clusters by simply adding these values together. For example, the 3 cohorts highlighted in the example below could be combined onto a single Exadata system using VM Clusters.

Cohorts	Metrics			Resulting Config					
	Allocated Storage (GB)	DB Servers Required	Storage Servers Required	DB Servers	OCPU Peak with OS+GI	OCPU Peak for Standby with OS+GI	Storage Servers	Total Storage (GB)	
AWS-US-EAST-2	18,582	0.31	0.45	2	17	16	3	153,60	
AWS-US-EAST-1	166,940	1.49	4.08	2	64	62	4	208,67	
EDC-North	73,149	0.98	1.79	2	26	19	3	153,60	
Unknown	840	0.04	0.02	2	5	5	3	153,60	
PL3-05 (Consolidated Lab)	168	0.02	0	2	5	5	3	153,60	
EDC-East	57,445	0.67	1.4	2	15	15	3	153,60	
unassigned	62,213	1.2	1.52	2	62	23	3	153,60	
DCEAST	377,916	2.79	9.23	3	77	77	9	472,35	
DCNORTH	471,255	4.33	11.51	5	218	154	12	589,06	

As shown above, these would easily fit into 2 Database Servers and 3 Storage Servers, while still allowing for full redundancy. The sum of DB Servers Required would be less than 2, so a 2nd DB Server could be used to provide redundancy in the event of a failure. Further consolidation onto Virtual Machines could be done if redundancy (High Availability) is not required for these cohorts.

Sub-Dividing Large Cohorts

In some cases, the defined cohorts are either too large for the target Exadata deployment model or might be larger than desired. In extremely large deployments (with hundreds or thousands of databases), it might be necessary or desirable to sub-divide cohorts and place them on multiple Exadata systems rather than a single large system.

OCCA does not currently provide guidance to users in sub-dividing large cohorts, so this must be done manually. OCCA users will need to determine how to sub-divide a single large cohort into multiple smaller cohorts and re-run the analysis starting at the cohort definition phase (Step #4).

Resource Consumption Plots

OCCA generates plots of aggregated resource consumption for each cohort of databases. These aggregations are time-aligned to account for the benefits of database consolidation. The primary metrics in these plots are:

- vCPU
- IOPS
- Memory (SGA & PGA)
- Storage space

Each plot includes the PEAK observed value over the time period of the capture, which is indicated by the RED line in the plot and printed at the top of the plot.

Annual Adjustment & Seasonality

Resource consumption analysis is performed using the previous ~33 days of hourly metrics in most cases. The exact number of days can vary slightly and is expressed in terms of hours (approximately 768 hours). The detailed analysis using hourly data is then adjusted (upwards only) if there is a prior period in the past year that showed higher consumption.

The adjustment plots give a view of how the adjustment was made, as well as a view of seasonality in the consumption metrics. Seasonality can be critical to capacity planning for enterprises that experience wide swings in resource consumption throughout the year.

Resource Consumption Percentiles

OCCA Analytics calculates resource consumption percentiles and includes these in the consumption plots. These reference lines (**blue dashed lines**) are intended for discussion purposes in the OCPU management plan (see section below).

Adjusted vCPU Consumption

OCCA generates both non-adjusted and adjusted plots on the desktop. The non-adjusted versions can be tracked back to metrics that reside in Enterprise Manager and were extracted from Enterprise Manager. The adjusted metrics are used for sizing because they account for the highest peak seen annually. The following plots should be used for sizing and customer presentations:

- `occa_sizing_output/plots/adjusted/<cohort_name>/<format>/`
 - `<cohort_name>_DB_vCPU_for_Standby_adjusted.<format>`
- `occa_sizing_output/plots/adjusted/<cohort_name>/<format>/`
 - `<cohort_name>_Adjustments.<format>`

The format of files specified by `<format>` will be PNG images or HTML with embedded Plotly graphs. The PNG files are static images, whereas HTML/Plotly provide more interactive analysis.

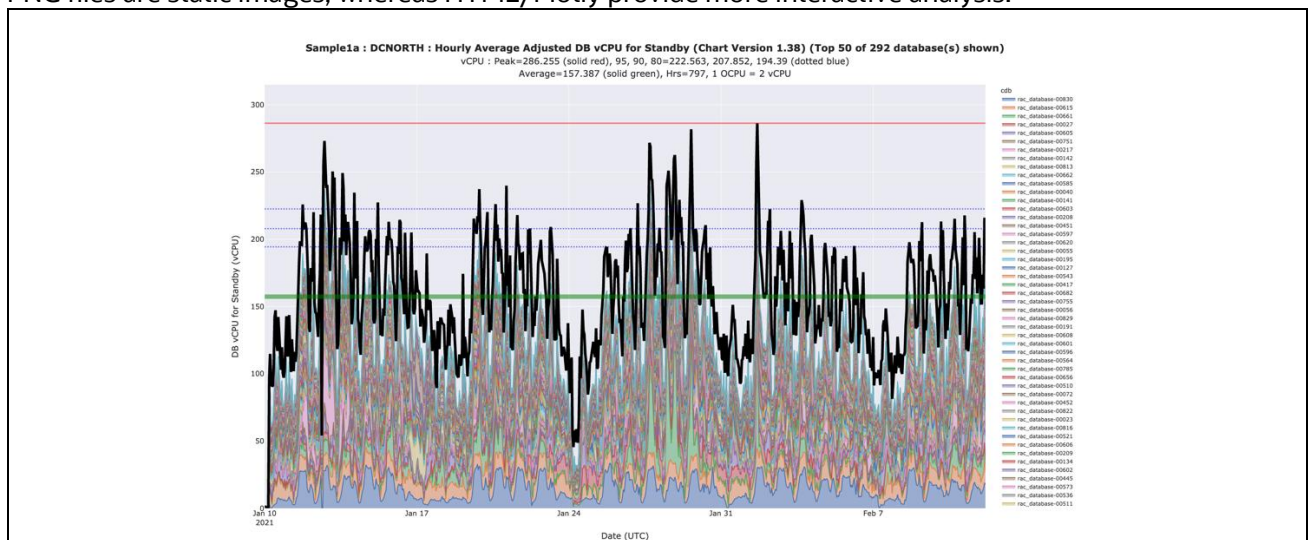


Figure 3: Adjusted vCPU for Standby

OCCA Analytics generates vCPU plots for standby (normal operation) as well as the case where all Standby databases become activated (Disaster condition). As noted previously, vCPU calculations are based on consumption of vCPU by databases and do not include O/S and GI resource consumption.

Adjustments & Seasonality

The adjustments plot shows how adjustments were calculated based on seasonal fluctuations over the prior 12 months. In addition to showing how adjustments were made, this plot can also be used to help the

customer determine how to best manage their systems based on any seasonal fluctuations throughout the year.

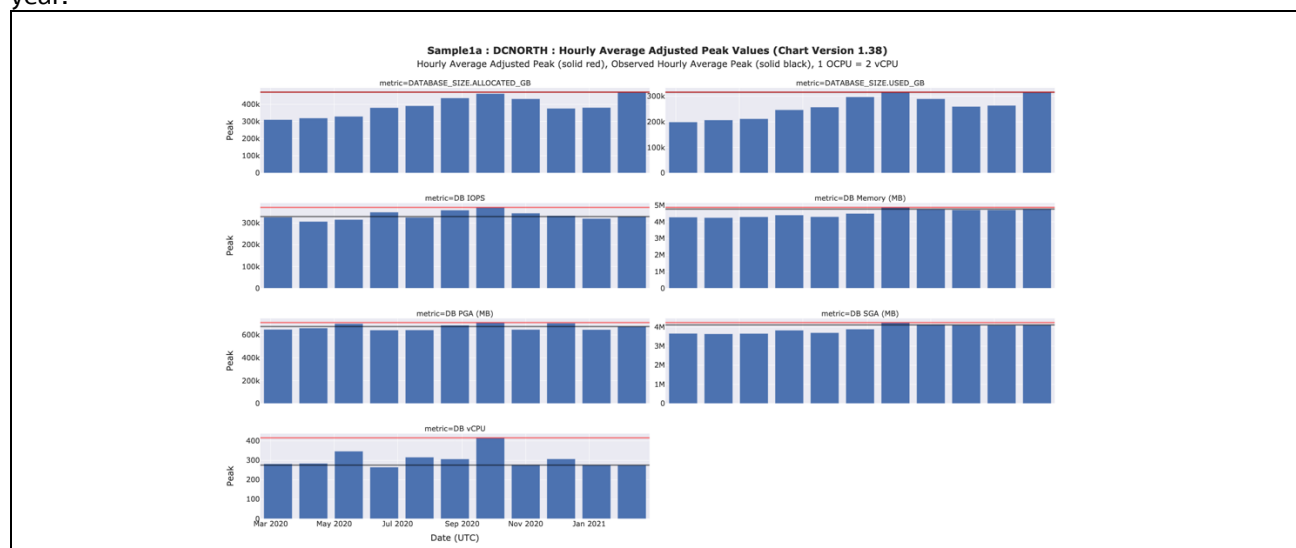


Figure 4: Adjustments Plot

OCPU Management Plan

Based on analysis of the resource consumption graphs and discussions with the customer, we put together an OCPU management plan that the customer will implement.

- OCPU Static "Set & Forget" Approach
- OCPU Month-to-Month Modifications
- OCPU Day-to-Day and Week-to-Week Modifications
- OCPU Dynamic Scaling

OCPU Static "Set & Forget" Approach

For customers looking for a simple "set & forget" approach to OCPU management, the RED line in the plots is the highest point observed over the past 12 months.

OCPU Month-to-Month Modifications

For customers with predictable seasonable trends, the month-to-month adjustments chart shows how vCPU consumption varied over the past 12 months.

OCPU Day-to-Day and Week-to-Week Modifications

For customer with predictable trends on a day-to-day or week-to-week basis, the preferred approach might be to manage OCPU settings according to the hourly detail plots. For example, a customer might have DEV/TEST systems that are not used on weekends, so the OCPU settings can be reduced dramatically during weekends. Another customer might have database cohorts that are busy only during month-end processing.

OCPU Dynamic Scaling

The vCPU consumption that would be seen with Dynamic Scaling in ExaDB-C@C and ExaDB-D (documented in MOS note# [2719916.1](#)) will be higher than the green line due to differences in timing. Dynamic Scaling uses a 2-hour time interval, whereas EMCC data is based on a 1-hour time interval.

The vCPU plot and monthly adjustments plot give information that is useful for developing the OCPU management plan for a customer. Using a simple "set & forget" approach will reflect the highest OCPU billing charges based on observed resource consumption over the past 12 months.

Growth & Fault Tolerance

OCCA does not automatically calculate sizing for growth or fault tolerance on platforms that allow Database Consolidation including Exadata, ExaDB-C@C, or ExaDB-D (users can select the HA (fault tolerance) option for BaseDB). Growth and fault tolerance should be handled at the aggregate level for each cohort by ensuring there is sufficient capacity.

Growth

OCCA does not automatically calculate growth. However, growth trends can be seen in the month-by-month adjustment data that shows multipliers relative to the latest month of detailed hourly metrics. Figure 4 shows an example of month-by-month data and trends experienced by each cohort.

It is not possible to determine growth trends by simply looking at these charts. Experience in managing the systems is required to understand what happened during the capture period. For example, what appears to be "growth" in these charts could simply mean databases were moved under control of Enterprise Manager.

Fault Tolerance

As with growth, notice that our end-state physical sizing (in the Merged Environments table) also provides sufficient excess capacity to provide fault tolerance. We recommend to first size based on current state resource consumption, then evaluate whether there is sufficient excess physical capacity to accommodate growth and provide fault tolerance. The customer needs to be acutely aware of fault tolerance (and growth) capabilities of the system, so it's important to not incorporate these requirements too early in the process.

Disaster Recovery

OCCA gives visibility to sizing for the overall resource pool needed by each cohort, as well as the resource needs during normal operation. Fault tolerance should be considered in light of both HA and DR needs.

Adding Capacity for Cohorts

OCCA allows users to add Database Servers to each cohort (or for ALL cohorts) to provide sufficient room for both fault tolerance and growth.

OCCA Desktop Analytics Parameter Settings

The OCCA Analytics code includes a number of settings that can be customized to change behavior of the code and control outputs. The following parameters are available:

Parameter	Default	Description
CREATE_OBFUSCATED_FILES=True/False	False	Controls spreadsheet file production
CREATE_SIMULATOR_FILES_ONLY=True/False	False	Controls spreadsheet file production
COHORTS=['<cohort_name>']	Off	Can be used to perform analysis only for specific cohorts
PLOT_ADJUSTMENT_MULTIPLIERS=True/False	True	Enable or disable creation of Adjustment Multiplier Plots
PLOT_SUM_OF_MAX=True/False	False	Controls printing of black dashed line that shows Sum of Max calculations
PLOT_INSTANCE_COUNT_LIMIT=250	250	Limits printing of detail when instance count exceeds this threshold (reduces memory usage)
PLOT_HTML=True/False	True	Controls printing of Plots in HTML files (interactive Plotly output)
PLOT_PNG=True/False	False	Controls printing of Plots in PNG image files
PLOT_PERCENTILES=True/False	True	Enables or disables creation of percentile calculations
PLOT_SUPPRESSED_COHORTS=['<cohort_name>']	off	Suppress the plots for cohorts that lead to Plotly crashes due to desktop DRAM limits.

PLOT_UNADJUSTED_VALUES=True/False	False	Determine whether to plot unadjusted values
PRECISION=3	3	Number of digits to which to round calculated fields
TIME_PERIODS=['03', '06', '09', '12']	'03','06','09','12'	Control which daily time periods are used to adjust hourly averages
OVERLAP_DAYS_WITHIN_A_COHORT = 7	7	For AWR Miner, set minimum days overlap between .out files in the same Cohort, reduce only when needed.

By default, the parameter settings are taken from a global copy of the parameter file located in the "occa" directory. It is possible to modify these parameters globally for all sizings run using that set of source code. It is also possible to use settings for a specific opportunity by copying the parameter file to the working directory. The search path for the parameter file (occa_Sizing_Parameters.py) is as follows:

1. Working Directory
2. /Users/<username>/Library/Python/3.11/lib/python/site-packages/occa/

In many cases, the parameters should only be set for a specific customer by copying the parameter file to the working directory.

Using OCCA AWR Miner

OCCA AWR Miner is the latest version of the AWR Miner tool that collects data from AWR History of individual Oracle Databases. OCCA AWR Miner is integrated with the EMCC Extracts scripts and OCCA as outlined in this document.

OCCA AWR Miner Components

OCCA AWR Miner consists of the following components:

AWR Miner Components	
Customer Extract	extracts/AWR-Miner-Readme.[pdf txt]
	extracts/awr_miner.sql
Oracle Internal	occa --awr-easy
	occa --awr-parse
	occa --awr-plot
	occa --awr-run
	occa --awr-import

Components for customers are found in the "extracts" directory that is sent to customers. Components for Oracle personnel to run internally are part of the "occa" executable. Customers provide ".out" files for each of their databases. The directory structure for OCCA AWR Miner is as follows:

Directory Structure	
<work_dir>/	Customer Extract Working Directory
awr_miner_out	Place 1 or more OCCA AWR Miner ".out" files in this directory. Files must use the original naming convention generated by the OCCA AWR Miner output scripts. Modifying file names may cause errors.
awr_miner_parsed	This automatically generated directory will contain all data parsed from the ".out" files.
awr_miner_plots	This automatically generated directory will contain the traditional AWR Miner plots for tuning and analysis, but not used for Sizing.
awr_miner_occa	This automatically generated directory will contain all plots and .csv files created during AWR Metric

Analysis. This directory will also contain the JSON file used for upload into OCCA Web.

OCCA AWR Miner Sizing Steps

1. Customer executes OCCA AWR Miner Scripts
2. Move .out files into the **awr_miner_out** directory
3. Parse AWR Miner Data: **occa --awr-parse .**
4. Generate OCCA AWR Miner Plots: **occa --awr-plot**
5. Copy Original Property Files, for example:

```
copy awr_miner_occa/properties/awr_tz_database_original.csv
    awr_miner_occa/properties/awr_tz_database.csv
copy awr_miner_occa/properties/awr_properties_database_original.csv
    awr_miner_occa/properties/awr_properties_database.csv
```
6. Edit Properties
 - a. Set time zone for each database in `awr_tz_database.csv`
 - b. Set cohort for each database in `awr_properties_database.csv`
7. Run AWR Metric Analysis: **occa --awr-run**
8. Sanity check results
 - a. Check contents of `awr_miner_occa/sizing/databases.csv` (look for exclusions)
 - b. Check contents of `awr_miner_occa/sizing/instances.csv` (look for exclusions)
 - c. Check Snapshot Coverage Overlap of AWR Miner extracts
See: `awr_miner_occa/plot/awr_snap_coverage_all_cohorts.html`
9. Generate JSON File for Upload: **occa --awr-import**
10. Upload JSON file (named "awr_miner_occa/awr_occa_upload_data.json") into OCCA Web

Customer Executes OCCA AWR Miner Scripts

The OCCA AWR Miner script (`awr_miner.sql`) and corresponding readme file (`AWR-Miner-Readme.pdf`) are found in the EMCC Extracts directory. If the customer is only using OCCA AWR Miner and not EMCC Extracts, there is an `AWR-Miner.zip` file that can be shared with the customer.

If the customer is capturing AWR Miner data for multiple databases that will be combined into a cohort, those AWR Miner extracts should be run approximately on the same day or within a short window not longer than approximately 2 days apart. The overlap of snapshots is important for consolidation sizing, especially when customers use the default AWR retention setting of 8 days.

AWR Miner extracts should cover the period of peak business processing.

AWR Snapshot Intervals & Retention

In general, larger volumes of data result in more accurate sizing. AWR lacks the months of history found in Enterprise Manager but can still be useful for sizing. It's not often feasible to change AWR settings and wait weeks or months for results, but it's important to understand what is available.

AWR Snapshot intervals longer than 1 hour are not usable for sizing. The default snapshot interval of 1 hour is acceptable, but shorter intervals such as 30-, 20-, 15-, or 10-minutes produce even better results. The shortest interval supported by Oracle Database is 10-minutes. OCCA will reject any AWR Miner extracts with longer than 1 hour snapshot intervals. The customer can modify the snapshot interval and wait for the duration of the retention period (8 days by default) for AWR data to accumulate.

The default AWR data retention is 8 days, but longer retention periods are preferred. Changing the retention period will require waiting for AWR data to accumulate, which isn't often feasible during a sizing

engagement. Longer AWR retention produces better results because it covers a longer time period that captures more workload and longer business cycles. For example, month-end processing won't be captured during an 8-day period if AWR extract is run in the middle of the month.

Time Zone Handling

Proper time alignment of database resource consumption is critical when databases are combined in a cohort. OCCA AWR Miner reconciles consumption metrics for databases by converting snapshot interval times to UTC (Universal Time Coordinated).

Older versions of AWR Miner extract files may work with OCCA but lack time zone information for each database. Users must manually supply time zone information for each database when an older version of the AWR Miner extract script is used. It is critical to enter the CORRECT time zone of each database because the offset is critical for analysis of simultaneous resource consumption.

OCCA AWR Miner extracts include time zone offset, which is used to determine the proper time zone. However, the NAME of the time zone might not be appropriate for a given customer, such as northern hemisphere vs. southern hemisphere or across continents. Databases in Europe and Africa or North America and South America might reside in the same time zone, but the time zone name can be important for customer confidence.

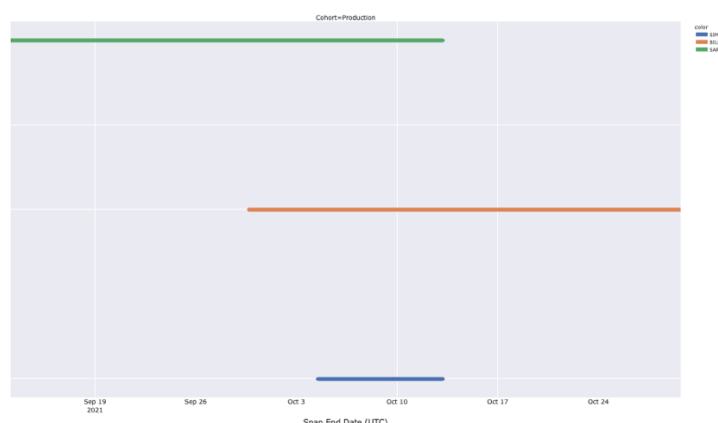
The "occa --awr-plot" command generates a list of valid values that can be used to enter or modify time zone information into a properties file.

- Valid Values: `awr_miner_occa/awr_miner_tz_choices.csv`
- Enter Properties: `awr_miner_occa/properties/awr_tz_database.csv`

Any valid value is acceptable and accurate, as long as the time zone offset is correct. For example, Vancouver, Los Angeles, and Tijuana all reside in the same time zone, but these cities reside in 3 different countries. Using the proper time zone name is important and this cannot be done automatically.

AWR Snapshot Overlap

Overlap of AWR snapshots is critical for databases combined into a single cohort. OCCA provides a chart to help visualize the overlap of AWR snapshot periods to determine whether we have the proper data to analyze. The following example shows 3 databases, where 2 of them had AWR data extracts performed on the same day and the AWR extract for 1 was performed much later. One of these databases (the BLUE line) had a much shorter AWR retention period.

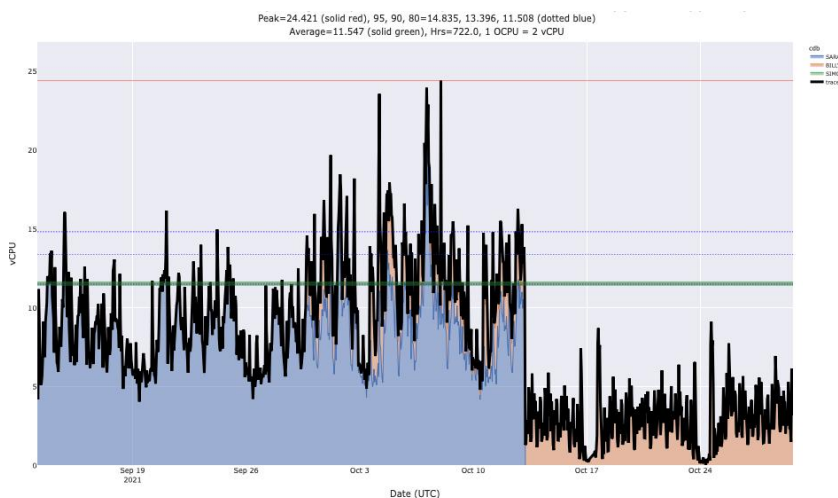


File: `awr_snap_coverage_all_cohorts.html`

These databases have some amount of overlap in the AWR snapshot intervals (at the beginning of October), but the different retention periods will affect the metrics as seen in the cohort charts consumption plots.

OCCA AWR Miner Cohorts

AWR Miner in OCCA implements the same cohort concept used for EMCC data in OCCA. Cohorts are set in the database properties file (properties_database.csv). OCCA generates plots for each defined cohort and the databases included with each cohort.



File: <cohort_name>_DB_vCPU_unadjusted.html

This example shows a steep drop in vCPU consumption between October 10th and 17th due to the day/time when the AWR Miner extracts were run. The **GREEN** database in this case has extremely low vCPU consumption and does not have much impact on the consumption overall. Note that Color coding of the Snap Coverage plot matches color coding of the vCPU consumption plot, which is the most commonly used plot. Other plots are sorted differently, so color coding doesn't match across all cases.

OCCA AWR Miner Database Plots

The OCCA AWR Miner database plots are designed to highlight the primary resource consumption metrics that are important to sizing for each individual database. OCCA AWR Miner will give a high-level view of performance that can be used to characterize performance of existing systems, but often does not contain sufficient data for details performance diagnosis. OCCA AWR Miner extracts can be useful in performance analysis by providing visibility to time periods where more detailed analysis using full AWR reports would be helpful.

Handling Non-US Numeric Separators

If AWR ".out" files were produced in a location that uses non-US numeric separators, the following solution may be required. The following environment variable swaps comma (',') and period ('.') characters that are present in the extract files.

Note: The OCCA AWR Miner version 25.2.0 and higher sets the appropriate NLS parameters so that the report produced will no longer have this issue.

On Linux/MacOS, run the following command:

```
export AWR_MINER_ADJUST_NUMERIC_SEP=Y
```

On Windows, run the following command

```
set AWR_MINER_ADJUST_NUMERIC_SEP=Y
```

Set this environment variable before running the **occa --awr-parse** commands to correct issues with non-US numeric separators. In the case where some AWR Miner files have US numeric separators and some have non-US numeric separators, the files can be parsed separately, and the resulting parsed files can be placed into a single directory to run **occa --awr-plot**.

OCCA Web Manual Data Entry

OCCA Web allows users to create opportunities manually rather than automatically creating opportunities by uploading JSON files from OCCA Analytics. Auto-uploaded opportunities cannot be modified in OCCA

Web (aside from creating scenarios and selecting targets), but manually created opportunities can be modified.

- Create New Opportunity from Manage Opportunities page
- Create Cohorts
- Add Databases to Cohorts
- Add Servers and Map Databases to them

OCCA Web uses APEX Interactive Grids, which allow direct data entry. Simply click on the "edit" button to begin editing data in each Interactive Grid. After each database is entered, the link for Server Management will appear. Each database must be mapped to at least 1 server or an error will result.

Sum of Peaks Method

For any manually entered Sizing Opportunity, OCCA will use the "sum of peaks" method. The "Sizing Method" column of the Exadata Sizing Output report will show "Sizing by SUM of DB Consumption Metrics" was used rather than the time-aligned peaks that are calculated in Cohort Rollups. Manually entered data includes a SINGLE data point for each database and resource type (vCPU, Memory, Storage, and I/O), so these metrics lack the necessary time-series data.

The Sum of Peaks method typically over-sizes database systems, especially with large numbers of databases consolidated onto a single system.

Debugging OCCA Issues

OCCA provides several features to assist in debugging issues that might occur during EMCC Extract processing, during OCCA Analytics (Python) execution, and on the OCCA Web side (APEX) processing.

Collecting Sizing Artifacts

Developers often need copies of sizing artifacts (extracts and properties files) in order to diagnose issues with OCCA sizing processes. The following command generates a ZIP file for sharing with the OCCA development team.

- Run the command: **occa --create-artifacts**

The zip file generated by this command is placed in the current working directory. The artifact zip file can be sent to OCCA Developers for analysis.

Debugging for Data Guard Handling

Execute the command "occa --run-metric-analysis" with "-d" command with a comma-separated list of database names.

Evaluating Server Utilization vs. DB Consumption

Utilization of server CPU resources and consumption of CPU by databases on the server are closely related and often need to be evaluated together. The volume of data in the extracts is typically too large for analysis using Microsoft Excel or other spreadsheet tools. The following command reduces the hourly metrics to facilitate analysis:

- Run the command: **occa -reduce-hourly-metrics**

This command generates a file named "reduced_hourly_metrics.csv" that can be analyzed using Excel. The OCCA Analytics package contains a sample spreadsheet named "Visualize_reduced_hourly_metrics.xlsx" that can be used to analyze server CPU utilization vs. database CPU consumption. Simply open this spreadsheet and use the Data→Refresh function to analyze the data.

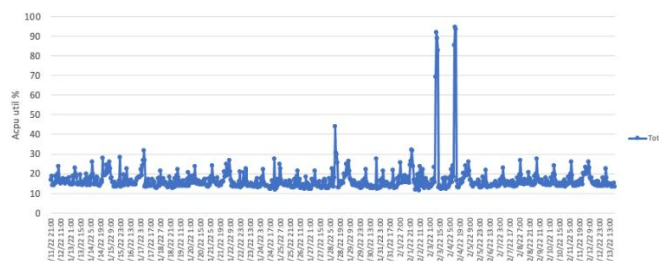
Evaluating Attenuated DB vCPU Outliers

OCCA (version 1.14.0 and later) will automatically attenuate DB vCPU consumption outliers that can arise on high thread-count servers. The amount of vCPU (logical CPU, hyper-threads, etc.) consumed by the

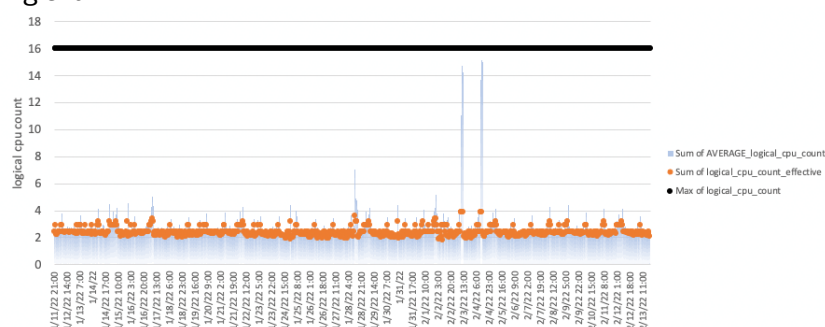
database can be inflated on certain combinations of Operating System and server hardware when CPU utilization is high. In cases where OCCA attenuates vCPU metrics, OCCA Analytics produces a file that includes details for analysis as well as an Excel spreadsheet to visualize the results.

- host_effective_cpu_metrics.csv
- Visualize_host_effective_cpu_metrics.xlsx

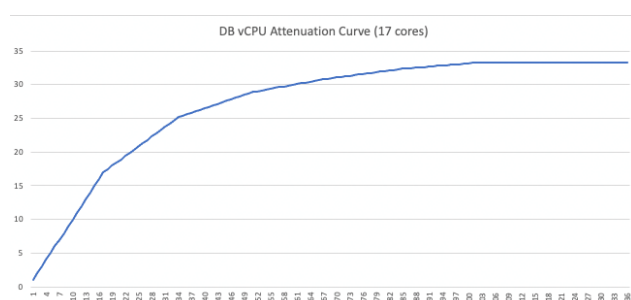
Simply open the spreadsheet and use the Data→Refresh function in Excel to determine whether hosts have been under heavy load leading to inflated vCPU metrics and how OCCA attenuated those values. The following chart shows a host under heavy CPU utilization.



In the periods where host CPU utilization goes high, the vCPU consumption reported to the Oracle Database by the Operating System becomes inflated. This inflation is a progressive curve, so OCCA cannot simply divide all vCPU metrics by a single value. OCCA includes the logic to attenuate these inflated values as shown in the following chart:

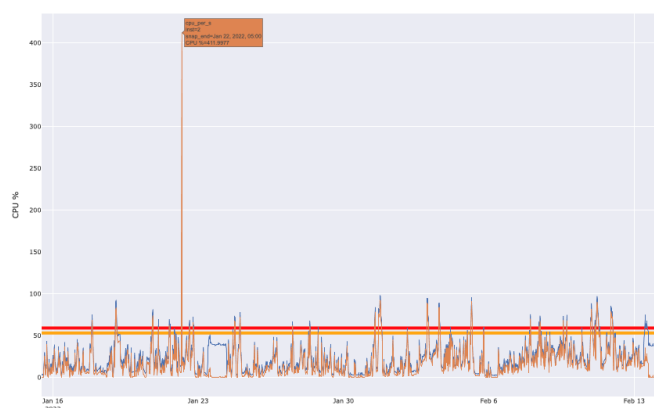


Inflated values are shown in light blue, whereas the effective (attenuated) vCPU metrics are shown in orange. The degree of attenuation rises progressively as the corresponding host CPU utilization increases. The following example shows the attenuation curve for a host with 17 cores.



Automatic Elimination of CPU Outliers

OCCA automatically eliminates outlier CPU utilization values that are clear measurement errors. CPU utilization exceeding 100% is not possible, but sometimes appears in the extracted data. OCCA automatically identifies these erroneous values and discards them. The following chart shows CPU utilization of 411%, which is a clear measurement error.



OCCA Command Reference

OCCA Analytics has been integrated into a single "occa" command rather than plain-text scripts, which handles all of the Python3 dependencies automatically using the pip3 installer and does not require Anaconda.

OCCA COMMAND	DESCRIPTION
occa --h, --help	Show expanded help
occa --create-properties	Create Properties Files for editing
occa --copy-db-name	Copy DB Name and ALL metrics from DG PRIMARY to STANDBY
occa --add-properties	Add Properties into processing stream
occa --run-metric-analysis	Run OCCA Metric Analysis
occa --emcc-import	Create JSON file for input into OCCA Web
occa --version	Version of OCCA package
occa --guide	Shows user's guide information
occa --path	Path to installed OCCA scripts (path might be hidden by O/S)
occa --copy-db-size	Deprecated. Copy DB Size (only) from DG PRIMARY to STANDBY
occa --copy-sizing	Copy completed EMCC sizing properties to OCCA properties
occa --copy-spreadsheets	Copy static spreadsheets to local directory
occa --create-artifacts	Create a ZIP file with all artifacts needed for developers to debug
occa --pre-flight	Run pre-flight checks and analysis
occa --reduce-db-metrics	Reduce DB Metrics (not normally needed)
occa --reduce-hourly-metrics	Reduce Hourly Metrics (not normally needed)
occa --reports	View reports through an HTML page
occa --awr-easy	Run Parse and Plot in a single command
occa --awr-parse	Parse OCCA AWR Miner Output Files
occa --awr-plot	Plot OCCA AWR Miner data after parsing
occa --awr-run	Run sizing analysis on AWR Miner data
occa --awr-import	Create JSON file for input into OCCA Web
occa --awr-reports	View reports through an HTML page
occa --checkmetrics	Optional flag for run-metric-analysis
occa --params	Optional flag to show execution parameters

Conclusion

Sizing is a collaborative effort between Oracle and the customer that relies on accurate input from the customer to deliver an accurate sizing estimate.

References

OCCA User's Guide for Oracle Employees

<https://oracle.sharepoint.com/:b:/t/ExadataPM/EZlCgndiZ4IAktQlONvxayoBWlbS69WN325K-2GsDRATbg?e=uXyz3k>

EMCC Extracts Guide for Customers

<https://oracle.sharepoint.com/sites/dbsg/Team-Assets/Shared-Assets/Utilities/DB-Sizing/EMCC-Extracts-Guide.pdf>

Scripts for Customer Extract

<https://occa.us.oracle.com/ords/r/occa/occa/occa-releases>

OCCA Analytics Desktop Package for Oracle Employees

<https://occa.us.oracle.com/ords/r/occa/occa/occa-releases>

OCCA Test Data (EMCC Extract)

https://objectstorage.uk-london-1.oraclecloud.com/OCCA_test_data_Sample3b_EMCC.zip

OCCA Web User Interface

<https://occa.us.oracle.com/>

Introduction to OCCA Presentation

<https://oracle.sharepoint.com/:p:/t/ExadataPM/ESghYSEppopBqOz8IX7WCGwBKMxtXyay4Xhi6Zs7EfxiVA?e=RYGxaH>

Pricing Calculators

<https://www.oracle.com/cloud/cost-estimator.html>